UP-PCS PRELIMS 2026 · HIGH PROBABILITY QUESTION BANK

Master Expected Questions — Part 3

High-Yield Questions 151 to 225 (Random Subject Distribution with UPPSC Traps & Ratta Points)

QUESTION RANGE
Questions 151 – 225

PATTERN & STYLE
UPPSC 2024–2025 Standard (Statements, A&R, Match Lists)

STUDY FORMAT
Question → Options → Answer → In-Depth Ratta & Trap Notes

TARGET DATE
December 6 Prelims Examination

Q. 151 📜 Modern Indian History 📌 Lord Dalhousie & Doctrine of Lapse Chronology

Arrange the following Indian Princely States annexed by Lord Dalhousie under the 'Doctrine of Lapse' in chronological order:

1. Satara
2. Jhansi
3. Sambalpur
4. Nagpur

Select the correct answer from the codes given below:

A 1 — 3 — 2 — 4

B 1 — 2 — 3 — 4

C 3 — 1 — 2 — 4

D 1 — 3 — 4 — 2

✓ **Correct Answer :Option A**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- Lord Dalhousie (1848–1856) applied the aggressive imperialist policy of **Doctrine of Lapse**, refusing to recognize adopted heirs to sovereign princely thrones:
    1. **Satara (1848):** First state annexed under the Doctrine.
    2. **Jaitpur (Bundelkhand) & Sambalpur (Odisha):** Annexed in **1849**.
    3. **Baghat (Himachal):** Annexed in **1850** (later restored by Lord Canning).
    4. **Udaipur (Chhattisgarh):** Annexed in **1852** (later restored).
    5. **Jhansi:** Annexed in **1853** (after Gangadhar Rao died without natural heir).
    6. **Nagpur:** Annexed in **1854** (Bhonsle state; royal jewels were publicly auctioned in Calcutta).

⚠️ **आयोग का TRAP / RATTA POINT**

Awadh was NOT annexed under the Doctrine of Lapse! Awadh was annexed in **February 1856 on charges of "misgovernance and maladministration"** based on James Outram's report!

Q. 152 🏛️ Indian Polity 📌 Panchayati Raj & 73rd Constitutional Amendment Act

Consider the following statements regarding the 73rd Constitutional Amendment Act, 1992:

1. It added Part IX to the Constitution titled "The Panchayats" and the Eleventh Schedule containing 29 functional items.
2. Under Article 243D, not less than one-third of the total number of seats and offices of chairpersons are reserved for women.
3. The State Election Commission responsible for conducting panchayat elections is established under Article 243K.

Which of the statements given above are correct?

A 1 and 2 only

B 2 and 3 only

C 1 and 3 only

D 1, 2 and 3

✓ **Correct Answer :Option D**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
    - **73rd CAA 1992 (effective 24 April 1993 - National Panchayati Raj Day):** Added Part IX (Articles 243 to 243-O) and **11th Schedule containing 29 functional items**.
    - **Article 243D(3):** Mandates not less than $\frac{1}{3}$ (33%) reservation for women among seats and chairpersons (many states like Bihar, MP, Uttarakhand have raised this to 50%).
    - **Article 243K:** State Election Commissioner appointed by Governor, removed only in like manner as a High Court judge.
    - **Article 243I:** State Finance Commission constituted every 5 years by the Governor.

⚠️ **आयोग का TRAP / RATTA POINT**

11th Schedule (Panchayats) has **29 items**; 12th Schedule (Municipalities) has **18 items**! Gram Sabha is defined under Article 243(b) and established under Article 243A.

Q. 153 🌍 Geography of India 📌 Origins of the Indus River Tributaries

Match List-I (Indus Tributary) with List-II (Source of Origin) and select the correct answer:

List-I (River)	List-II (Source / Origin)
A Jhelum	1 Rakshastal Lake near Mansarovar (Tibet)
B Chenab	2 Verinag Spring at the foot of Pir Panjal (Kashmir)
C Ravi	3 Confluence of Chandra and Bhaga streams at Tandi (Himachal Pradesh)
D Sutlej	4 West of Rohtang Pass in Kullu Hills (Himachal Pradesh)

A A-2, B-3, C-4, D-1

B A-1, B-2, C-4, D-3

C A-2, B-4, C-3, D-1

D A-3, B-2, C-4, D-1

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Jhelum:** Originates from deep spring at Verinag; flows through Srinagar and Wular Lake before entering Pakistan through a deep gorge at Baramulla.
- **Chenab (Asikni):** Largest Indus tributary; formed by two headstreams, Chandra and Bhaga, joining at Tandi near Keylong in Lahaul (HP).
- **Ravi (Parushni):** Originates west of Rohtang Pass in the Kullu hills of HP; flows between Pir Panjal and Dhauladhar ranges.
- **Sutlej (Shutudri):** Antecedent trans-Himalayan river originating from Rakshastal Lake (Tibet); cuts through Zaskar range via Shipki La pass into India.

⚠️ आयोग का TRAP / RATTA POINT

Beas (Vipas) river originates near Rohtang Pass (Beas Kund) and **meets Sutlej entirely within India at Harike Wetland (Punjab)**!

Q. 154 🔬 General Science (Biology) 📌 Genetics & Nucleic Acids (DNA vs RNA)

Consider the following statements regarding nucleic acids:

1. James Watson and Francis Crick proposed the Double Helix model of DNA in 1953 based on X-ray diffraction data of Rosalind Franklin.
2. In DNA, the purine Adenine pairs with Thymine via two hydrogen bonds ($A=T$), while Guanine pairs with Cytosine via three hydrogen bonds ($G \equiv C$).
3. In RNA, the pyrimidine base Thymine is replaced by Uracil.

Which of the statements given above are correct?

A 1 and 2 only

B 2 and 3 only

C 1 and 3 only

D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Double Helix Model (1953):** Two antiparallel polynucleotide chains coiled around a central axis; Watson, Crick, and Maurice Wilkins shared the Nobel Prize in 1962.
  - **Base Pairing (Chargaff's Rule):** Purines = Pyrimidines ($A+G = T+C$). Adenine pairs with Thymine ($A=T$, two H-bonds); Guanine pairs with Cytosine ($G \equiv C$, three H-bonds).
  - **DNA vs RNA:** DNA contains 2-deoxyribose sugar and bases A, T, G, C. RNA contains ribose sugar and replaces **Thymine (T) with Uracil (U)**.

⚠️ आयोग का TRAP / RATTA POINT

Purines are double-ringed (**Adenine, Guanine**); Pyrimidines are single-ringed (**Cytosine, Thymine, Uracil**).

Q. 155 🎯 Uttar Pradesh Special 📌 Major Buddhist Heritage Sites in UP

Match List-I (Buddhist Site in UP) with List-II (Historical Significance) and select the correct answer:

List-I (Site)	List-II (Significance)
A Sarnath (Isipatana / Rishipattana)	1 Site of Buddha's *Mahaparinirvana* (death) under two Sal trees; Ramabhar Stupa
B Kushinagar (Kusinara)	2 First Sermon of Buddha (*Dhammacakkappavattana*); Dhamek Stupa & Ashoka Lion Capital
C Shravasti (Sahet-Mahet)	3 Capital of Kosala where Buddha spent maximum rainy retreats (*Vassavasa*); Jetavana Monastery
D Sankisa (Farrukhabad)	4 Legendary site where Buddha descended from Trayastrimsa Heaven via a golden ladder

A A-2, B-1, C-3, D-4

B A-1, B-2, C-3, D-4

C A-2, B-3, C-1, D-4

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Sarnath (Varanasi):** Buddha preached his first sermon (*Dharmachakra Pravartana*) to 5 disciples (Kaundinya, etc.); site of Dhamek and Chaukhandi stupas.
- **Kushinagar:** Buddha attained *Mahaparinirvana* in 483 BCE at age 80; site of colossal 6-meter reclining Buddha statue and Ramabhar Stupa (cremation ground).
- **Shravasti:** Buddha spent 24-25 rainy seasons (*Chaturmasa*); converted ruthless bandit Angulimala; wealthy merchant Anathapindika donated the Jetavana monastery.
- **Sankisa (Farrukhabad):** Buddhist tradition holds Buddha descended from heaven here accompanied by Indra and Brahma; site of Ashokan elephant capital pillar.

⚠️ आयोग का TRAP / RATTA POINT

Buddha was born in **Lumbini** (Nepal), attained Enlightenment at **Bodh Gaya** (Bihar), gave First Sermon at **Sarnath** (UP), and died at **Kushinagar** (UP)!

Q. 156 🌿 Environment & Ecology 📌 Ecological Concepts and Principles

Match List-I (Ecological Concept) with List-II (Definition / Key Feature) and select the correct answer:

List-I (Concept)	List-II (Definition)
A Ecotone	1 Functional role, resource utilization, and status of an organism in its ecosystem
B Ecological Niche	2 Zone of junction or transition between two or more diverse ecosystems (e.g. mangrove, marshland)
C Edge Effect	3 Phenomenon where species diversity or population density is greater in the ecotone than in flanking communities
D Keystone Species	4 Species whose presence and activity exerts disproportionately large influence on the entire community structure

A A-2, B-1, C-3, D-4

B A-1, B-2, C-3, D-4

C A-2, B-3, C-1, D-4

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Ecotone:** Transition zone between two biomes (e.g., mangrove between marine and terrestrial; grassland between forest and desert).
- **Ecological Niche (Joseph Grinnell 1917):** "Address" is habitat, but "profession" is niche; unique functional role and trophic position occupied by a species. Two species cannot occupy the exact same niche indefinitely (Gause's Competitive Exclusion Principle).
- **Edge Effect:** The heightened richness, density, and diversity of species in the ecotone boundary compared to adjacent communities.
- **Keystone Species (Robert Paine 1969):** Species that plays a critical role in maintaining ecological balance (e.g., Tiger in forest, Sea Otter in kelp forest, Fig tree in tropical rainforest).

⚠️ आयोग का TRAP / RATTA POINT

Difference between Habitat and Niche: Many species can share the same Habitat, but NO two species can occupy the exact same Niche in the long run!

Q. 157 Indian Economy | Reserve Bank of India (RBI) Structure & Evolution

Consider the following statements regarding the Reserve Bank of India:

1. The Reserve Bank of India was established on April 1, 1935, based on the recommendations of the Royal Commission on Indian Currency and Finance (Hilton Young Commission).
2. The RBI was initially set up as a private shareholders' bank and was nationalized on January 1, 1949.
3. Sir Osborne Smith was the first Governor of the RBI, and C.D. Deshmukh was the first Indian Governor.

Which of the statements given above are correct?

A 1 and 2 only	B 2 and 3 only
C 1 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Foundation:** Established under the Reserve Bank of India Act, 1934; commenced operations on April 1, 1935, in Calcutta (shifted permanently to Bombay in 1937).
  - **Nationalization:** Under the RBI (Transfer to Public Ownership) Act, 1948, RBI was fully nationalized with effect from **January 1, 1949**.
  - **Governors:**
  - 1st Governor = **Sir Osborne Smith** (1935–1937).
  - 2nd Governor = Sir James Braid Taylor (1937–1943).
  - 1st Indian Governor = **Sir Chintaman Dwarakanath (C.D.) Deshmukh** (1943–1949).

⚠ आयोग का **TRAP / RATTA POINT**

Dr. Manmohan Singh is the ONLY person in Indian history to have served as BOTH Governor of the RBI (1982–1985) and Prime Minister of India!

Q. 158 Current Affairs | Global Gender Gap Report 2024

Consider the following statements regarding the Global Gender Gap Report 2024 published by the World Economic Forum (WEF):

1. Iceland was ranked as the most gender-equal country in the world for the 15th consecutive year.
2. The index measures gender parity across four key dimensions: Economic Participation & Opportunity, Educational Attainment, Health & Survival, and Political Empowerment.
3. India was ranked 129th out of 146 economies in the 2024 edition.

Which of the statements given above are correct?

A 1 and 2 only	B 2 and 3 only
C 1 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **WEF Global Gender Gap Report 2024:** Published annually since 2006.
  - **Top Rank:** Iceland continues to lead the world with a score of 0.935 (closing over 93% of its gender gap), followed by Finland and Norway.
  - **India's Standing:** Ranked **129th out of 146 nations** (score 0.641). India scored well on Educational Attainment and Political Empowerment, but registered low scores on Economic Participation and Opportunity.

⚠ आयोग का **TRAP / RATTA POINT**

Published by **World Economic Forum (WEF)**, NOT UNDP or UN Women! (UNDP publishes Gender Inequality Index - GII).

Q. 159 📜 Ancient Indian History 📌 Indus Valley Civilization River Alignments

Match List-I (Harappan City) with List-II (River on Which It Was Situated) and select the correct answer:

List-I (City)	List-II (River)
A Harappa	1 Indus
B Mohenjodaro	2 Ravi
C Lothal	3 Sutlej
D Ropar (Rupnagar)	4 Bhogava

A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Harappa:** Situated on the left bank of **Ravi** river in Sahiwal (Montgomery) district, Punjab, Pakistan (excavated by Daya Ram Sahni 1921).
- **Mohenjodaro:** Situated on the right bank of **Indus** river in Larkana district, Sindh, Pakistan (excavated by Rakhaldas Banerjee 1922).
- **Lothal:** Situated along the **Bhogava** river (tributary of Sabarmati) near the Gulf of Khambhat in Gujarat.
- **Ropar (Rupnagar):** Situated on the left bank of the **Sutlej** river in Punjab (excavated by Y.D. Sharma; famous for human burial with dog).

⚠️ आयोग का TRAP / RATTA POINT

Common river matches to memorize: - Alamgirpur = **Hindon** (Meerut, UP) - Kalibangan = **Ghaggar** (Rajasthan) - Banawali = **Rangoi / Ghaggar** (Haryana) - Suktagendor = **Dasht** (Baluchistan).

Q. 160 🌍 World Geography 📌 Strategic Islands & Oceanic Locations

Match List-I (Strategic Island) with List-II (Ocean Basin / Governing Entity) and select the correct answer:

List-I (Island)	List-II (Location / Jurisdiction)
A Diego Garcia	1 South Atlantic Ocean; British Overseas Territory (claimed by Argentina)
B Falkland Islands (Malvinas)	2 Chagos Archipelago, Central Indian Ocean; major joint UK/US military base
C Reunion Island	3 Arabian Sea / Gulf of Aden; sovereign territory of Yemen
D Socotra Archipelago	4 Southwest Indian Ocean; Overseas Department of France

A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Diego Garcia:** Largest atoll in the Chagos Archipelago (Indian Ocean); highly strategic US-UK naval and bomber base. (In 2024, UK agreed to hand over sovereignty of Chagos Archipelago to Mauritius while securing 99-year lease for Diego Garcia base).
- **Falkland Islands:** Disputed between Argentina and UK; 1982 Falklands War; located in cold South Atlantic Ocean.
- **Reunion Island:** Volcanic island in southwest Indian Ocean east of Madagascar; integral territory of France.
- **Socotra Archipelago:** Located near the mouth of the Gulf of Aden; sovereign territory of Yemen; famous for unique endemic flora like the Dragon's Blood Tree (*Dracaena cinnabari*).

⚠️ आयोग का TRAP / RATTA POINT

Diego Garcia is in the INDIAN OCEAN, not the Pacific!

Q. 161 🏛 Indian Polity 📌 Right to Freedom of Religion (Articles 25 to 28)

Match List-I (Article of the Constitution) with List-II (Guaranteed Freedom) and select the correct answer:

List-I (Article)	List-II (Guaranteed Freedom)
A Article 25	1 Freedom from payment of taxes for promotion of any particular religion
B Article 26	2 Freedom of conscience and free profession, practice, and propagation of religion
C Article 27	3 Freedom from attending religious instruction or worship in state-funded educational institutions
D Article 28	4 Freedom to manage religious affairs and establish institutions for religious/charitable purposes

A A-2, B-4, C-1, D-3

B A-4, B-2, C-1, D-3

C A-2, B-1, C-4, D-3

D A-3, B-4, C-1, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Article 25:** Guarantees to ALL persons freedom of conscience and right to freely profess, practice, and propagate religion (individual right, subject to public order, morality, and health).
- **Article 26:** Guarantees to religious denominations the collective right to manage their own religious affairs and acquire property (collective denomination right).
- **Article 27:** State cannot compel any citizen to pay any TAX whose proceeds are specifically appropriated for promoting a particular religion (fees can be levied for administrative services, but not taxes).
- **Article 28:** Completely bans religious instruction in educational institutions wholly maintained out of State funds.

⚠ आयोग का TRAP / RATTA POINT

Article 25 protects INDIVIDUALS; Article 26 protects RELIGIOUS DENOMINATIONS/COLLECTIVES! Under Article 25 Explanation II, the reference to Hindus includes Sikhs, Jains, and Buddhists.

Q. 162 📜 Medieval Indian History 📌 Delhi Sultanate Fiscal Taxes

Match List-I (Fiscal Tax in Delhi Sultanate) with List-II (Nature / Levied Upon) and select the correct answer:

List-I (Tax)	List-II (Nature)
A Kharaj	1 Poll tax levied on non-Muslim protected subjects (*Zimmis*) in lieu of military service
B Jizya	2 Land revenue tax levied on agricultural produce (usually $\frac{1}{3}\text{rd}$ to $\frac{1}{2}$)
C Zakat	3 State's share ($\frac{1}{5}\text{th}$) of the spoils of war and booty
D Khams	4 Religious tax of 2.5% levied exclusively on well-off Muslims for charitable purposes

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Kharaj:** Agricultural land tax on cultivable lands (Alauddin Khalji raised Kharaj to 50% of gross produce based on measurement / *Masaht*).
- **Jizya:** Capitation/poll tax paid by non-Muslims; Firoz Shah Tughlaq was the first Sultan to levy Jizya on Brahmins as a separate tax.
- **Zakat:** Obligatory wealth tax (2.5%) payable only by wealthy Muslims, deposited in *Bait-ul-Mal* for poor relief.
- **Khams:** One-fifth ($\frac{1}{5}$ or 20%) of the war spoils (*Ghanimah*) went to the State, and $\frac{4}{5}$ was distributed among soldiers. (Alauddin Khalji and Muhammad bin Tughlaq reversed this rule: the State took $\frac{4}{5}$ and left only $\frac{1}{5}$ to soldiers!).

⚠ आयोग का TRAP / RATTA POINT

Sharb was an irrigation tax levied by **Firoz Shah Tughlaq** at the rate of 10% of produce from lands irrigated by royal canals!

Q. 163 🔬 General Science (Biology) 📌 Human Circulatory System & Heart Physiology

Consider the following statements regarding the human heart and blood vessels:

1. The Sinoatrial (SA) node located in the right atrium acts as the natural pacemaker of the heart.
2. Arteries always carry oxygenated blood away from the heart, with the sole exception of the Pulmonary Artery, which carries deoxygenated blood to the lungs.
3. In a standard Electrocardiogram (ECG), the P-wave represents the ventricular repolarization.

Which of the statements given above is/are correct?

A 1 and 2 only	**B** 2 and 3 only
C 1 only	**D** 1, 2 and 3

✓ **Correct Answer :Option A**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- Statements 1 and 2 are correct: The SA node generates rhythmic action potentials (~72 times/minute), initiating cardiac contraction. Arteries carry blood away from the heart at high pressure; the Pulmonary Artery is the ONLY artery carrying deoxygenated blood (from right ventricle to lungs). The Pulmonary Vein is the ONLY vein carrying oxygenated blood (from lungs to left atrium).
- Statement 3 is incorrect: In a standard ECG:
  - **P-wave:** Represents **Atrial Depolarization** (electrical excitation leading to contraction of both atria).
  - **QRS Complex:** Represents **Ventricular Depolarization** (contraction of ventricles).
  - **T-wave:** Represents **Ventricular Repolarization** (return of ventricles from excited to resting state).

⚠️ **आयोग का TRAP / RATTA POINT**

Tricuspid valve guards the right atrioventricular opening; Bicuspid (Mitral) valve guards the left atrioventricular opening!

Q. 164 🎯 Uttar Pradesh Special 📌 Tribes of Uttar Pradesh (Tharu & Buksa)

Consider the following statements regarding tribes in Uttar Pradesh:

1. Tharu is the most populous scheduled tribe in Uttar Pradesh, residing primarily in the Terai region of Maharajganj, Siddharthnagar, Shravasti, Bahraich, and Lakhimpur Kheri.
2. The Tharu tribe traditionally observes Diwali as a day of sorrow (*Shok Parva*).
3. The Buksa (Bhoksa) tribe resides primarily in Bijnor district of Uttar Pradesh.

Which of the statements given above are correct?

A 1 and 2 only	**B** 2 and 3 only
C 1 and 3 only	**D** 1, 2 and 3

✓ **Correct Answer :Option D**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Tharu Tribe:** Largest tribal group in UP; belongs to Kirat Mongoloid stock; practice Joint Family system; follow matriarchal traits; claim origin from Rajputana desert; celebrate Diwali as an occasion to mourn deceased ancestors (*Shok Parva*). A university for Tharus was established at Lakhimpur Kheri.
  - **Buksa (Bhoksa):** Found primarily in **Bijnor** district; claim descent from the Pawars of Dharanagar; social stratification mirrors Hindu caste hierarchy.
  - **Other UP Tribes:** Sahariya (Lalitpur), Agariya, Parahiya, Chero, Baiga (Sonbhadra).

⚠️ **आयोग का TRAP / RATTA POINT**

Highest tribal population in UP is in **Sonbhadra** district; districts with the lowest/virtually zero ST population are **Baghpat and Kannauj**!

Q. 165 🌿 Environment & Ecology 📌 Ramsar Convention & Montreux Record

Consider the following statements regarding the Montreux Record:

1. The Montreux Record is a register of wetland sites on the Ramsar List where ecological character has changed, is changing, or is likely to change due to human interference.
2. Currently, two Indian wetlands are listed on the Montreux Record: Keoladeo National Park (Rajasthan) and Loktak Lake (Manipur).
3. Chilika Lake (Odisha) was placed on the Montreux Record in 1993 but was successfully removed in 2002 after ecological restoration.

Which of the statements given above are correct?

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Montreux Record:** Established under Ramsar COP in 1990; highlights wetlands requiring urgent conservation priority.
  - **Current Indian Sites (2):** 1. **Keoladeo National Park (Rajasthan):** Listed due to water shortage and grazing imbalance. 2. **Loktak Lake (Manipur):** Listed due to deforestation in catchment and proliferation of *Phumdis* (floating vegetation mats) impacting Keibul Lamjao National Park.
  - **Chilika Lake:** India's first Ramsar site (along with Keoladeo, 1981); placed on Montreux in 1993 due to siltation; removed in 2002 (awarded the prestigious Ramsar Wetland Conservation Award).

⚠️ आयोग का TRAP / RATTA POINT

Keoladeo and Loktak REMAIN on the Montreux Record; Chilika Lake is NO LONGER on it!

Q. 166 📈 Indian Economy 📌 India's Foreign Trade Structure

As per recent Economic Survey and Ministry of Commerce data, which country was India's LARGEST trading partner (combined exports and imports) in the fiscal year 2023–24?

A. United States of America (USA)
B. China
C. United Arab Emirates (UAE)
D. Russia

✓ Correct Answer :Option B

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- In FY 2023–24, **China** emerged as India's largest two-way merchandise trading partner with total bilateral trade valued at **\$118.4 billion** (India's exports to China: \$16.67 billion; imports from China: \$101.75 billion, resulting in a massive trade deficit of \$85.08 billion).
- The **USA** was the second largest trading partner at **\$118.3 billion** (India enjoys a significant trade surplus of over \$35 billion with the USA).
- **Top 5 Trading Partners (2023–24):** 1. China, 2. USA, 3. UAE, 4. Russia, 5. Saudi Arabia.
- Russia is India's 4th largest trading partner primarily driven by discounted crude oil imports.

⚠️ आयोग का TRAP / RATTA POINT

India has the largest merchandise **trade SURPLUS with the USA**, and the largest merchandise **trade DEFICIT with China**!

Q. 167 Current Affairs | Global Events & Pageants 2024

The 71st edition of the Miss World pageant was hosted in Mumbai, India, in March 2024 (returning to India after 28 years). Who was crowned Miss World 2024?

A Krystyna Pyszkova (Czech Republic)	**B** Yasmina Zaytoun (Lebanon)
C Karolina Bielawska (Poland)	**D** Sini Shetty (India)

✓ **Correct Answer :Option A**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Krystyna Pyszkova** of the **Czech Republic** was crowned Miss World 2024 at the Jio World Convention Centre in Mumbai on March 9, 2024 (crowned by outgoing Miss World Karolina Bielawska of Poland).
- First Runner-up was Yasmina Zaytoun of Lebanon.
- Sini Shetty represented India and made it to the Top 8.
- India last hosted the Miss World pageant in Bangalore in 1996.

⚠ आयोग का TRAP / RATTA POINT

Six Indian women have won the Miss World title: Reita Faria (1966), Aishwarya Rai (1994), Diana Hayden (1997), Yukta Mookhey (1999), Priyanka Chopra (2000), and Manushi Chhillar (2017).

Q. 168 Modern Indian History | Constitutional Proposals of the 1940s Chronology

Arrange the following constitutional initiatives and missions sent by the British in chronological order:

1. August Offer
2. Cripps Mission
3. Wavell Plan (Shimla Conference)
4. Cabinet Mission

Select the correct answer from the codes given below:

A 1 — 2 — 3 — 4	**B** 1 — 3 — 2 — 4
C 2 — 1 — 3 — 4	**D** 1 — 2 — 4 — 3

✓ **Correct Answer :Option A**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

1. **August Offer (8 August 1940):** Announced by Viceroy Lord Linlithgow; proposed Dominion status after the war and an expanded Governor-General's Executive Council; recognized Indians' right to frame their own constitution for the first time.
2. **Cripps Mission (March–April 1942):** Led by Sir Stafford Cripps; offered Dominion status and a Constituent Assembly *after* WWII, with option for provinces to secede; rejected by Congress (Gandhi called it a "post-dated cheque on a failing bank").
3. **Wavell Plan & Shimla Conference (June–July 1945):** Convened by Lord Wavell; broke down over Jinnah's demand that the Muslim League alone could nominate all Muslim members to the Executive Council.
4. **Cabinet Mission (March–May 1946):** Comprising Lord Pethick-Lawrence, Sir Stafford Cripps, and A.V. Alexander; rejected Pakistan; proposed a 3-tier federation (Groups A, B, C) and laid the framework for the Constituent Assembly.

⚠ आयोग का TRAP / RATTA POINT

Chronology sequence: **August Offer (1940) -> Cripps (1942) -> C.R. Formula (1944) -> Wavell Plan (1945) -> Cabinet Mission (1946) -> Mountbatten Plan (3 June 1947)**!

Q. 169 🌍 Geography of India 📌 Multipurpose River Valley Projects & Dams

Match List-I (Dam / Project) with List-II (River & State) and select the correct answer:

List-I (Dam / Project)	List-II (River & State)
A Tehri Dam	1 Sutlej River; Himachal Pradesh & Punjab
B Bhakra-Nangal Project	2 Bhagirathi River; Uttarakhand
C Hirakud Dam	3 Narmada River; Gujarat
D Sardar Sarovar Dam	4 Mahanadi River; Odisha

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Tehri Dam:** Highest dam in India (260.5 m); rock and earth-fill embankment dam constructed at the confluence of Bhagirathi and Bhilangna rivers in Tehri, Uttarakhand.
- **Bhakra Dam:** One of the highest straight gravity dams in the world (226 m) on Sutlej River; reservoir is Govind Sagar.
- **Hirakud Dam:** Longest earthen dam in the world (main dam 4.8 km, total length with dykes 25.8 km) on Mahanadi River in Sambalpur, Odisha.
- **Sardar Sarovar Dam:** Concrete gravity dam on Narmada River at Kevadiya, Gujarat (powers Gujarat, MP, Maharashtra, Rajasthan).

⚠️ आयोग का TRAP / RATTA POINT

Tehri is the HIGHEST dam in India (260.5 m); Hirakud is the LONGEST dam in India (main dyke 4.8 km); Bhakra is the highest GRAVITY dam!

Q. 170 🏛️ Indian Polity 📌 Parliamentary Devices & Motions

Match List-I (Parliamentary Motion) with List-II (Objective / Requirement) and select the correct answer:

List-I (Motion)	List-II (Objective / Requirement)
A Adjournment Motion	1 Requires support of 50 members; moved only in Lok Sabha against the entire Council of Ministers to test its majority
B No-Confidence Motion	2 Moved to draw attention of the House to an urgent matter of definite public importance; requires support of 50 members
C Calling Attention Motion	3 Indian parliamentary innovation (since 1954) to call minister's attention to urgent public matter and seek authoritative statement
D Censure Motion	4 Can be moved against individual ministers or entire cabinet for specific policies; if passed, Council of Ministers need not resign

A A-2, B-1, C-3, D-4

B A-1, B-2, C-3, D-4

C A-2, B-3, C-1, D-4

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Adjournment Motion:** Extraordinary device interrupting normal business; requires support of at least 50 members to be admitted; discussions last not less than 2.5 hours; contains an element of censure against government.
- **No-Confidence Motion (Rule 198 of Lok Sabha):** Moved ONLY in Lok Sabha; requires support of 50 members; need not state specific reasons; if passed, the entire government must resign.
- **Calling Attention Motion:** Unique Indian parliamentary invention (1954); like Zero Hour, not mentioned in standard Westminster rules.
- **Censure Motion:** Must specify grounds for censure; moved against individual ministers or entire council; its passage does NOT mandate resignation of the government.

⚠️ आयोग का TRAP / RATTA POINT

Both "Zero Hour" and "Calling Attention Motion" are indigenous Indian parliamentary innovations! Zero Hour started in 1962; Calling Attention in 1954.

Q. 171 📜 Ancient Indian History 📌 Six Schools of Classical Indian Philosophy (Shad-Darshana)

Match List-I (Philosophical School) with List-II (Founder / Chief Propounder) and select the correct answer:

List-I (Darshana)	List-II (Founder)
A Samkhya	1 Sage Gautama
B Nyaya	2 Sage Kapila
C Vaisheshika	3 Sage Jaimini
D Purva Mimamsa	4 Sage Kanada (Uluka)

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- The 6 Orthodox (*Astika*) Schools of Vedic Philosophy:
  - **Samkhya:** Founded by **Sage Kapila** (*Samkhya Sutra*); oldest system; propounds dualism of *Purusha* (consciousness) and *Prakriti* (matter).
  - **Yoga:** Founded by **Patanjali** (*Yoga Sutra*); practical path of Ashtanga yoga.
  - **Nyaya:** Founded by **Sage Gautama** (*Nyaya Sutra*); school of logic and epistemology (4 Pramanas: Pratyaksha, Anumana, Upamana, Shabda).
  - **Vaisheshika:** Founded by **Sage Kanada** (*Vaisheshika Sutra*); atomic theory of the universe (*Paramanu*).
  - **Purva Mimamsa:** Founded by **Sage Jaimini** (*Mimamsa Sutra*); emphasizes ritualism and authority of the Vedas.
  - **Uttara Mimamsa (Vedanta):** Founded by **Badarayana** (*Brahma Sutra*).

⚠️ आयोग का TRAP / RATTA POINT

Mimamsa was founded by Jaimini; Vedanta (Uttara Mimamsa) was founded by Badarayana!

Q. 172 🔬 General Science (Biology) 📌 Mineral Deficiencies and Human Disorders

Match List-I (Mineral Nutrient) with List-II (Deficiency Manifestation) and select the correct answer:

List-I (Mineral)	List-II (Manifestation)
A Iodine	1 Microcytic Hypochromic Anemia
B Iron	2 Goitre (enlargement of thyroid gland) & Cretinism in infants
C Fluorine (in trace deficiency vs excess)	3 Rickets, Osteoporosis, and muscle tetany
D Calcium	4 Tooth decay (deficiency < 1 ppm) & Dental/Skeletal Fluorosis (excess > 1.5–2 ppm)

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Iodine:** Essential component of thyroxine (T_4) and triiodothyronine (T_3); deficiency causes endemic goitre and stunted physical/mental growth in infants (Cretinism).
- **Iron:** Central constituent of hemoglobin; deficiency impairs oxygen transport, causing Nutritional Anemia.
- **Fluorine:** Required at ~1 ppm to strengthen tooth enamel (forming fluoroapatite); deficiency causes dental caries; excess (> 1.5–2 mg/L) causes mottled teeth (**Dental Fluorosis**) and bone deformities (**Skeletal Fluorosis / Knock-Knee syndrome**).
- **Calcium:** Major mineral in bones and teeth; required for blood clotting, nerve conduction, and muscle contraction; deficiency causes rickets, osteopenia, and tetany.

⚠️ आयोग का TRAP / RATTA POINT

Itai-Itai (ouch-ouch) disease is caused by **Cadmium (Cd)** poisoning; Minamata disease is caused by **Methyl Mercury (Hg)** poisoning!

Q. 173 🎯 Uttar Pradesh Special 📌 Power Infrastructure & Plants in UP

Match List-I (Power Plant in UP) with List-II (Type / District) and select the correct answer:

List-I (Plant)	List-II (District & Type)
A Narora Atomic Power Station (NAPS)	1 Sonbhadra (Pipri); Hydro-electric
B Obra Thermal Power Plant	2 Bulandshahr; Nuclear Power (PHWR)
C Rihand Hydro-Electric Project (Govind Ballabh Pant Sagar)	3 Sonbhadra (Shaktinagar); NTPC Coal Thermal
D Singrauli Super Thermal Power Station	4 Sonbhadra (Obra); UP Rajya Vidyut Utpadan Nigam Coal Thermal

A A-2, B-4, C-1, D-3

B A-4, B-2, C-1, D-3

C A-2, B-1, C-4, D-3

D A-3, B-4, C-1, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Narora Atomic Power Station (NAPS):** Located on the banks of the Ganges in **Bulandshahr** district; operates 2 Pressurized Heavy Water Reactors (PHWRs, 220 MW each). It is the ONLY nuclear power plant in Uttar Pradesh.
- **Obra Thermal Power Station:** Located on the Renu river in **Sonbhadra** district (built with Russian collaboration).
- **Rihand Dam Project:** Built on the Rihand river (tributary of Son) at Pipri in **Sonbhadra**; its reservoir, **Govind Ballabh Pant Sagar**, is India's largest artificial reservoir by volume.
- **Singrauli Thermal Power Station:** India's first NTPC super thermal coal plant, located at Shaktinagar in **Sonbhadra** ("Energy Capital of India").

⚠️ आयोग का TRAP / RATTA POINT

Sonbhadra district is known as the "Power Capital of India" (*Urja Anchala*) due to its immense concentration of thermal and hydro power stations!

Q. 174 🌿 Environment & Ecology 📌 Kigali Amendment to the Montreal Protocol

Consider the following statements regarding the Kigali Amendment to the Montreal Protocol:

1. It was adopted in October 2016 in Kigali, Rwanda, at the 28th Meeting of the Parties (MOP28).
2. It aims to phase down the production and consumption of Hydrofluorocarbons (HFCs), which are potent greenhouse gases.
3. Unlike CFCs, Hydrofluorocarbons (HFCs) do not deplete the stratospheric ozone layer, but they have high Global Warming Potentials (GWPs).

Which of the statements given above are correct?

A 1 and 2 only

B 2 and 3 only

C 1 and 3 only

D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Kigali Amendment (2016):** Legally binding amendment to Montreal Protocol; entered into force on 1 January 2019.
  - **Target Substance:** Phases down **Hydrofluorocarbons (HFCs)** by more than 80% over the next 30 years (expected to avoid up to 0.4°C of global warming by 2100).
  - **Chemical Nature:** HFCs were introduced as ozone-safe substitutes for CFCs and HCFCs because HFCs contain NO chlorine or bromine, hence having **Zero Ozone Depletion Potential (ODP)**. However, their Global Warming Potential is up to 12,000 times greater than CO_2.

⚠️ आयोग का TRAP / RATTA POINT

India belongs to Group 2 of developing countries under Kigali, committing to start phasing down HFCs in **2028** and reduce them by 85% by **2047**!

Q. 175 📈 Indian Economy 📌 FRBM Act & N.K. Singh Committee Recommendations

Consider the following statements regarding the Fiscal Responsibility and Budget Management (FRBM) Review Committee (headed by N.K. Singh):

1. The committee recommended adopting 'Debt-to-GDP ratio' as the primary operational fiscal anchor instead of relying solely on fiscal deficit.
2. It recommended a targeted general government debt-to-GDP ratio of 60% by FY 2023 (40% for the Central Government and 20% for State Governments).
3. It proposed an "Escape Clause" allowing the government to deviate from fiscal targets up to 0.5% of GDP in extraordinary circumstances such as war, national calamity, or severe economic crisis.

Which of the statements given above are correct?

A 1 and 2 only	B 2 and 3 only
C 1 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - The N.K. Singh Committee (appointed in 2016, submitted report in 2017):
  - **Debt Anchor:** 60% general government debt-to-GDP ratio (**40% for Centre, 20% for States**).
  - **Fiscal Deficit Target:** Central fiscal deficit to be gradually brought down to 2.5% of GDP.
  - **Escape Clause:** Flexibility to exceed the target by 0.5% of GDP on grounds of national security, acts of war, agricultural collapse, or structural reforms with unanticipated fiscal implications (incorporated into amended FRBM Act 2018).

⚠️ आयोग का TRAP / RATTA POINT

Debt ceiling split: **40% Centre + 20% States = 60% Total General Government Debt**!

Q. 176 ⚡ Current Affairs 📌 Sports & Paris Paralympics 2024

Consider the following statements regarding India's historic performance at the Paris 2024 Paralympic Games:

1. India recorded its best-ever performance in Paralympic history, winning 29 medals (7 Gold, 9 Silver, and 13 Bronze).
2. Avani Lekhara became the first Indian woman to win two Paralympic Gold medals by defending her title in the Women's 10m Air Rifle Standing SH1 event.
3. Sumit Antil successfully defended his Paralympic Gold medal in the Men's Javelin Throw F64 category with a Games record throw.

Which of the statements given above are correct?

A 1 and 2 only	B 2 and 3 only
C 1 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Medal Haul:** India finished **18th in the medal tally** with 29 medals (7 Gold, 9 Silver, 13 Bronze), far surpassing Tokyo 2020 (19 medals).
  - **Gold Medalists (7):** 1. Avani Lekhara (Shooting - 10m Air Rifle) 2. Nitesh Kumar (Para-Badminton SL3) 3. Sumit Antil (Para-Athletics - Javelin Throw F64, 70.59 m) 4. Harvinder Singh (Para-Archery - Men's Recurve Individual; first-ever archery gold for India) 5. Dharambir (Men's Club Throw F51) 6. Pravin Kumar (Men's High Jump T64) 7. Navdeep Singh (Men's Javelin Throw F41)

⚠️ आयोग का TRAP / RATTA POINT

Harvinder Singh won India's FIRST EVER Gold medal in archery at either the Olympics or Paralympics!

Q. 177 Medieval Indian History | Mughal Historical Literature & Authors

Match List-I (Mughal Historical Text) with List-II (Author) and select the correct answer:

List-I (Text)	List-II (Author)
A Humayun-Nama	1 Abbas Khan Sarwani
B Tarikh-i-Sher Shahi	2 Gulbadan Begum (Humayun's sister)
C Akbarnama & Ain-i-Akbari	3 Abdul Qadir Badauni
D Muntakhab-ut-Tawarikh	4 Abul Fazl

A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Humayun-Nama:** Written in Persian by **Gulbadan Begum**, daughter of Babur and sister of Humayun, at the request of Akbar; provides intimate glimpses of the Mughal royal harem.
- **Tarikh-i-Sher Shahi (Tuhfat-i-Akbar Shahi):** Composed by **Abbas Khan Sarwani** by order of Akbar; provides the most detailed contemporary account of Sher Shah Suri's administration.
- **Akbarnama:** Written by court historian **Abul Fazl** in three volumes (Volume 3 is the celebrated *Ain-i-Akbari*, a statistical compendium of Akbar's empire).
- **Muntakhab-ut-Tawarikh:** Written by orthodox critic **Abdul Qadir Badauni**; famous for severe criticisms of Akbar's liberal religious policies and *Din-i-Ilahi*.

⚠ आयोग का TRAP / RATTA POINT
Tuzuk-i-Baburi (Baburnama) was written by Babur himself in **Chagatai Turki**, later translated into Persian by Abdur Rahim Khan-i-Khanan under Akbar!

Q. 178 World Geography | Mediterranean Biome and Climate

Which of the following characteristics is/are associated with the Mediterranean Climate (Cs type in Koppen classification)?

1. Dry, warm summers and mild, rainy winters.
2. Predominance of westerly winds during the winter season bringing rainfall.
3. Sclerophyllous vegetation with thick waxy leaves and extensive deep root systems.
4. Viticulture (grape cultivation) and citrus fruit orchards form the backbone of agriculture.

Select the correct answer from the codes given below:

A. 1, 2 and 4 only
B. 1, 3 and 4 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

✓ Correct Answer :Option D

DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All four statements are correct:
  - **Distribution:** Western margins of continents between **30° and 45° latitudes** (Mediterranean basin, Central California, Central Chile, Cape Town region in South Africa, Southern and Southwestern Australia).
  - **Precipitation:** In summer, trade winds blow offshore (completely dry). In winter, shifting pressure belts bring moist on-shore **Westerlies**, causing winter rains.
  - **Vegetation:** Sclerophyllous scrub (*Maquis* in France, *Chaparral* in California, *Matorral* in Chile); stunted trees with thick bark, waxy leathery leaves to resist summer transpiration.
  - **Economy:** "Orchard lands of the World"; world capital of citrus fruits (oranges, lemons), olives, and commercial wine-making (*Viticulture*).

⚠ आयोग का TRAP / RATTA POINT
The defining diagnostic feature of Mediterranean climate: **RAIN IN WINTER, DROUGHT IN SUMMER**!

Q. 179 🏛️ Indian Polity 📌 Constitutional vs Non-Constitutional Bodies

Which of the following bodies is a NON-CONSTITUTIONAL (Statutory or Executive) body?

1. Central Vigilance Commission (CVC)
2. National Human Rights Commission (NHRC)
3. National Commission for Scheduled Tribes (NCST)
4. Lokpal and Lokayuktas

Select the correct answer from the codes given below:

A 1, 2 and 4 only	**B** 2, 3 and 4 only
C 1 and 3 only	**D** 1, 2, 3 and 4

✓ **Correct Answer :Option A**

📘 **DETAILED SOLUTION & HIGH-YIELD ANALYSIS**

- **Constitutional Bodies:** Explicitly provided in the Constitution:
  - National Commission for Scheduled Tribes (NCST) = **Article 338A** (inserted by 89th CAA 2003).
  - NCSC (Article 338), NCBC (Article 338B), Finance Commission (Article 280), Election Commission (Article 324), CAG (Article 148), UPSC (Article 315).
- **Statutory Bodies (Created by Acts of Parliament):**
  - **CVC:** Statutory status granted under Central Vigilance Commission Act, 2003 (originally set up 1964 via executive resolution on Santhanam Committee report).
  - **NHRC:** Created under Protection of Human Rights Act, 1993.
  - **Lokpal:** Created under Lokpal and Lokayuktas Act, 2013.

⚠️ **आयोग का TRAP / RATTA POINT**

National Commission for Women (NCW) and National Commission for Minorities (NCM) are STATUTORY bodies, NOT Constitutional bodies!

Q. 180 🔬 General Science (Physics) 📌 Electromagnetic Spectrum

Arrange the following electromagnetic waves in increasing order of their WAVELENGTH (from shortest to longest wavelength):

1. X-rays
2. Visible Light
3. Infrared Rays
4. Microwaves

Select the correct answer from the codes given below:

A 1 — 2 — 3 — 4	**B** 2 — 1 — 3 — 4
C 1 — 3 — 2 — 4	**D** 4 — 3 — 2 — 1

✓ **Correct Answer :Option A**

📘 **DETAILED SOLUTION & HIGH-YIELD ANALYSIS**

- In the Electromagnetic Spectrum, energy and frequency are inversely proportional to wavelength ($E = h\nu = \frac{hc}{\lambda}$):
  - **Gamma rays:** Shortest wavelength ($\lambda < 10^{-12} \text{ m}$), highest frequency/energy.
  - **X-rays:** $\lambda \approx 10^{-10} \text{ m}$ (discovered by W.C. Roentgen).
  - **Ultraviolet (UV):** $\lambda \approx 10^{-8} \text{ m}$.
  - **Visible light:** $\lambda \approx 400 \text{ nm}$ (violet) to 700 nm (red).
  - **Infrared (IR):** $\lambda \approx 10^{-5} \text{ m}$ (heat radiation; used in night-vision and TV remotes).
  - **Microwaves:** $\lambda \approx 10^{-2} \text{ m}$ (used in RADAR and microwave ovens).
  - **Radio waves:** Longest wavelength ($\lambda > 1 \text{ m}$), lowest frequency/energy.

⚠️ **आयोग का TRAP / RATTA POINT**

Increasing order of Wavelength: **Gamma < X-rays < UV < Visible < Infrared < Microwaves < Radio waves**! (Energy and frequency follow the exact opposite order).

Q. 181 🎯 Uttar Pradesh Special 📌 Scheduled Castes (SC) Demographics in UP

Consider the following statements regarding the Scheduled Caste (SC) population in Uttar Pradesh as per Census 2011:

1. Uttar Pradesh has the largest Scheduled Caste population in absolute numbers among all states in India.
2. Sitapur district has the highest Scheduled Caste population in absolute numbers in Uttar Pradesh.
3. Kaushambi district has the highest percentage proportion of Scheduled Caste population to its total population in the state.

Which of the statements given above are correct?

A 1 and 2 only
B 2 and 3 only
C 1 and 3 only
D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **State Total:** UP's SC population is **4.13 crore** (accounting for 20.69% of UP's total population and 20.5% of India's total SC population).
  - **Highest Absolute SC Population: Sitapur**, followed by Prayagraj (Allahabad) and Hardoi.
  - **Lowest Absolute SC Population: Baghpat**, followed by Shravasti and Gautam Buddha Nagar.
  - **Highest SC Percentage: Kaushambi (34.72%)**, followed by Machhlishahr/Jaunpur and Sitapur.
  - **Lowest SC Percentage: Baghpat (11.44%)**, followed by Bareilly and Balrampur.

⚠️ आयोग का TRAP / RATTA POINT
Absolute SC highest = **Sitapur**; Percentage SC highest = **Kaushambi**!

Q. 182 🌿 Environment & Ecology 📌 Ecological Terminology and Coining Scientists

Match List-I (Ecological Term) with List-II (Scientist Who Coined It) and select the correct answer:

List-I (Term)	List-II (Scientist)
A Ecology (*Oekologie*)	1 A.G. Tansley (1935)
B Ecosystem	2 Ernst Haeckel (1866)
C Biodiversity	3 Norman Myers (1988)
D Biodiversity Hotspot	4 Walter G. Rosen (1985)

A A-2, B-1, C-4, D-3
B A-1, B-2, C-4, D-3
C A-2, B-4, C-1, D-3
D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Ecology:** Coined by German zoologist **Ernst Haeckel** in 1866 (from Greek *oikos* meaning house/dwelling, and *logos* meaning study).
- **Ecosystem:** Coined by British botanist **Arthur G. Tansley** in 1935 as a basic structural and functional unit of nature.
- **Biodiversity (Biological Diversity):** Coined as a portmanteau by **Walter G. Rosen** in 1985 (though Raymond F. Dasmann used 'biological diversity' in 1968). E.O. Wilson popularized it.
- **Biodiversity Hotspots:** Coined by British ecologist **Norman Myers** in 1988 in his paper "The Environmentalist".

⚠️ आयोग का TRAP / RATTA POINT
Eugene P. Odum is known as the "Father of Modern Ecology"; Ramdeo Misra is revered as the "Father of Indian Ecology"!

Q. 183 📜 Modern Indian History 📌 Early Indian Newspapers & Editors

Match List-I (Historical Newspaper) with List-II (Founder / Editor) and select the correct answer:

List-I (Newspaper)	List-II (Founder / Editor)
A Bengal Gazette (Hicky's Gazette, 1780)	1 Raja Ram Mohan Roy (First Persian weekly)
B Mirat-ul-Akhbar (1822)	2 James Augustus Hicky (First printed newspaper in India)
C Rast Goftar (1851)	3 Ishwar Chandra Vidyasagar & Dwarkanath Vidyabhusan
D Som Prakash (1858)	4 Dadabhai Naoroji (Gujarati fortnightly)

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Hicky's Bengal Gazette (1780, Calcutta):** First printed newspaper in India; confiscated by British authorities for criticizing Warren Hastings.
- **Mirat-ul-Akhbar (1822):** Founded by Raja Ram Mohan Roy in Persian; closed down in 1823 in protest against John Adam's Licensing Regulations.
- **Rast Goftar ("The Truth Teller", 1851, Bombay):** Founded by Dadabhai Naoroji and Naoroji Furdunji to promote socio-religious reform among Parsis.
- **Som Prakash (1858):** Founded by Ishwar Chandra Vidyasagar and edited by Dwarkanath Vidyabhusan; actively championed the cause of indigo ryots; Lord Lytton's Vernacular Press Act 1878 was targeted primarily against it.

⚠️ आयोग का TRAP / RATTA POINT

Amrita Bazar Patrika (founded by Sisir Kumar Ghosh and Motilal Ghosh in 1868) morphed overnight from a Bengali-English weekly into an ENGLISH daily to escape the Vernacular Press Act 1878!

Q. 184 📈 Indian Economy 📌 Measures of Money Supply (M_1, M_2, M_3, M_4)

Match List-I (Money Aggregate) with List-II (Components Included) and select the correct answer:

List-I (Aggregate)	List-II (Components)
A M_1 (Narrow Money)	1 M_1 + Savings deposits with Post Office savings banks
B M_2	2 Currency with public + Demand deposits with banking system + 'Other' deposits with RBI
C M_3 (Broad Money)	3 M_3 + Total deposits with Post Office savings organisations (excluding NSC)
D M_4	4 M_1 + Time deposits (fixed deposits) with the banking system

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- Formulated by the Second Working Group on Money Supply (1977):
  - **M_1 (Narrow Money):** Currency notes/coins held by public + Demand Deposits (current/savings) of commercial banks + Other deposits with RBI. Most liquid.
  - **M_2:** M_1 + Post Office savings bank deposits.
  - **M_3 (Broad Money):** M_1 + Net Time Deposits of commercial banks. Most widely used measure of money supply in India.
  - **M_4:** M_3 + Total Post Office deposits (excluding National Savings Certificates - NSC). Least liquid.

⚠️ आयोग का TRAP / RATTA POINT

Decreasing order of Liquidity: **$M_1 > M_2 > M_3 > M_4$**! High Powered Money / Reserve Money is designated as **M_0**.

Q. 185 Current Affairs | International Heads & Organizations 2024

Consider the following pairs regarding heads of major international organizations:

1. President of the World Bank: Ajay Banga
2. Managing Director of IMF: Kristalina Georgieva
3. Director-General of World Trade Organization (WTO): Ngozi Okonjo-Iweala
4. President of the 79th session of UN General Assembly (UNGA): Philemon Yang (Cameroon)

How many of the above pairs are correctly matched?

A. Only two pairs
B. Only three pairs
C. All four pairs
D. Only one pair

✓ Correct Answer :Option C

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All four pairs are accurately matched:
  - **Ajay Banga:** Assumed office as the 14th President of the World Bank in June 2023 (first Indian-American to head World Bank).
  - **Kristalina Georgieva (Bulgaria):** Re-appointed for a second 5-year term as Managing Director of the International Monetary Fund (IMF) starting October 2024.
  - **Dr. Ngozi Okonjo-Iweala (Nigeria):** First woman and first African to serve as Director-General of the WTO.
  - **Philemon Yang:** Former Prime Minister of Cameroon, elected President of the 79th Session of the United Nations General Assembly (UNGA) in September 2024 (succeeding Dennis Francis of Trinidad and Tobago).

⚠️ आयोग का TRAP / RATTA POINT

Antonio Guterres (Portugal) is the Secretary-General of the United Nations (serving his second term since 2017).

Q. 186 Ancient Indian History | Harappan Burial Practices

Match List-I (Harappan Site) with List-II (Specific Burial Feature) and select the correct answer:

List-I (Site)	List-II (Burial Feature)
A. Lothal	1. Wooden coffin burials and Cemetery H & R-37
B. Harappa	2. Joint / Double burials (male and female skeleton interred together in single grave)
C. Ropar (Rupnagar)	3. Circular pit burials (pot burials) and brick-lined graves
D. Kalibangan	4. Human buried along with a domesticated dog in oval pit

A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Lothal:** Three double burials were discovered where two skeletons (male and female) were interred together in brick-lined graves (some scholars speculated early sati, though unsupported).
- **Harappa:** Evidence of **wooden coffin burials** (rosewood coffin) and two distinct cemeteries: Cemetery R-37 (mature phase) and Cemetery H (post-urban phase).
- **Ropar (Punjab):** Unique burial of a dog buried beneath a human master, reminiscent of Neolithic Burzahom (Kashmir).
- **Kalibangan:** Three types of disposal: extended inhumation, pot burials (fractional), and symbolic rectangular pits without skeletal remains.

⚠️ आयोग का TRAP / RATTA POINT

Burzahom (Kashmir Neolithic) and Ropar (Harappan) are the two classic sites showing **dog buried with human**!

Q. 187 🌍 Geography of India 📌 Channels and Water Passages of India

Match List-I (Channel / Water Passage) with List-II (Islands Separated) and select the correct answer:

List-I (Passage)	List-II (Separation)
A 8 Degree (8° N) Channel	1 Minicoy Island and Main Lakshadweep Archipelago
B 9 Degree (9° N) Channel	2 Minicoy Island (India) and Maldives
C 10 Degree (10° N) Channel	3 Little Andaman and South Andaman (Rutland Island)
D Duncan Passage	4 Little Andaman and Car Nicobar

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **8° N Channel:** Separates **Minicoy** (southernmost island of Lakshadweep, India) from the **Maldives**.
- **9° N Channel:** Separates **Minicoy** from the main **Lakshadweep** group (Kalpeni, Suheli, Kavaratti).
- **10° N Channel:** Separates the **Andaman group** (Little Andaman) from the **Nicobar group** (Car Nicobar) in the Bay of Bengal (approx. 150 km wide).
- **Duncan Passage:** Separates **South Andaman (Rutland Island)** from **Little Andaman**.

⚠️ आयोग का TRAP / RATTA POINT

Grand Channel separates **Great Nicobar** from Sumatra (Indonesia)!

Q. 188 🏛 Indian Polity 📌 Landmark Judgments on Article 21

Match List-I (Supreme Court Landmark Case) with List-II (Right Enunciated under Article 21) and select the correct answer:

List-I (Landmark Case)	List-II (Right Enunciated)
A Justice K.S. Puttaswamy v. Union of India (2017)	1 Right to Livelihood is an integral part of the Right to Life
B Olga Tellis v. Bombay Municipal Corporation (1985)	2 Fundamental Right to Privacy declared intrinsic to life and personal liberty
C Maneka Gandhi v. Union of India (1978)	3 Right to life includes the right to the enjoyment of pollution-free water and air
D Subhash Kumar v. State of Bihar (1991)	4 "Procedure established by law" must be just, fair, and reasonable (Due Process)

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **K.S. Puttaswamy (2017):** Unanimous 9-judge bench ruled that the **Right to Privacy** is a fundamental right protected under Article 21 and Part III.
- **Olga Tellis (1985 - "Pavement Dwellers Case"):** 5-judge bench led by CJI Y.V. Chandrachud ruled that the **Right to Livelihood** is an inescapable facet of the Right to Life.
- **Maneka Gandhi (1978):** Overruled *A.K. Gopalan (1950)*; imported the American doctrine of "substantive due process" into Article 21, ruling that procedure must be "just, fair, and reasonable".
- **Subhash Kumar (1991):** Held that Right to Life includes the **right to a pollution-free environment**.

⚠️ आयोग का TRAP / RATTA POINT

Right to Education was declared a fundamental right under Article 21 in *Mohini Jain (1992)* and *Unni Krishnan (1993)*, before the 86th CAA 2002 inserted Article 21A!

Q. 189 🔬 General Science (Chemistry) 📌 Ores and Extraction of Metals

Match List-I (Metal) with List-II (Principal Mineral Ore) and select the correct answer:

List-I (Metal)	List-II (Ore)
A Aluminium	1 Galena (PbS)
B Lead	2 Bauxite ($Al_2O_3 \cdot 2H_2O$)
C Mercury	3 Monazite sand
D Thorium	4 Cinnabar (HgS)

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ **Correct Answer :Option A**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Aluminium:** Extracted from **Bauxite** via the Bayer process and Hall-Héroult electrolytic reduction (cryolite Na_3AlF_6 is added to lower the melting point).
- **Lead:** Extracted from sulphide ore **Galena (PbS)**.
- **Mercury:** Extracted from sulphide ore **Cinnabar (HgS)** by roasting in air.
- **Thorium:** Radioactive actinide extracted from **Monazite beach sands** (plentiful in coastal Kerala and Odisha).

⚠️ **आयोग का TRAP / RATTA POINT**

Pitchblende is the principal ore of **Uranium**; Haematite and Magnetite are ores of **Iron**; Chalcopyrite and Malachite are ores of **Copper**!

Q. 190 🎯 Uttar Pradesh Special 📌 One District One Product (ODOP) Combinations

Match List-I (District of UP) with List-II (ODOP Product) and select the correct answer:

List-I (District)	List-II (ODOP Product)
A Aligarh	1 Sports Goods
B Meerut	2 Locks and Hardware
C Hathras	3 Amla (Indian Gooseberry) Food Processing
D Pratapgarh	4 Asafoetida (Hing) processing

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ **Correct Answer :Option A**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Aligarh:** Famous for Locks and brass hardware (*Tala Nagri*).
- **Meerut:** Premier sports goods manufacturing cluster in India (cricket bats, balls, sports equipment).
- **Hathras:** Known for Asafoetida (*Hing*) compounding and processing.
- **Pratapgarh:** Largest producer of Amla (Gooseberry) in India; famous for amla murabba, juice, and candies.

⚠️ **आयोग का TRAP / RATTA POINT**

Muzaffarnagar = Jaggery (*Gur*); Sambhal = Horn and bone craft; Rampur = Applique patchwork and Mentha!

Q. 191 🌿 Environment & Ecology 📌 Kunming-Montreal Global Biodiversity Framework (GBF)

Consider the following statements regarding the Kunming-Montreal Global Biodiversity Framework (GBF) adopted at COP15 to the Convention on Biological Diversity (CBD):

1. It adopted the landmark "30x30 target", committing nations to conserve at least 30% of global land, inland waters, and coastal and marine areas by 2030.
2. It aims to reduce global food waste by half and significantly reduce overconsumption and waste generation by 2030.
3. The Global Biodiversity Framework replaces the previous Aichi Biodiversity Targets (2011–2020).

Which of the statements given above are correct?

A 1 and 2 only	B 2 and 3 only
C 1 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Adoption:** Finalized in December 2022 in Montreal, Canada, under the presidency of China.
  - **Core Targets (23 global targets for 2030):**
  - **Target 3 ("30 by 30"):** Protect at least 30% of terrestrial, inland water, and coastal/marine areas by 2030.
  - **Target 16:** Halve global food waste and substantially reduce waste generation.
  - **Target 19:** Mobilize at least \$200 billion per year from all sources for biodiversity financing.
  - Replaces the 20 Aichi Biodiversity Targets adopted in Nagoya in 2010.

⚠️ आयोग का TRAP / RATTA POINT

CBD has 3 main objectives: Conservation of biodiversity, Sustainable use of components, and Fair and equitable sharing of benefits arising from genetic resources (Nagoya Protocol)!

Q. 192 📜 Medieval Indian History 📌 Delhi Sultanate Architectural Monuments

Match List-I (Monument) with List-II (Builder Sultan) and select the correct answer:

List-I (Monument)	List-II (Builder)
A Adhai Din Ka Jhonpra (Ajmer)	1 Alauddin Khalji
B Alai Darwaza (Delhi)	2 Qutbuddin Aibak
C Hauz Khas & Siri Fort	3 Ghiyasuddin Tughlaq
D Tughlaqabad Fort	4 Alauddin Khalji
	Note: Match carefully knowing Alai Darwaza and Siri Fort belong to Alauddin Khalji.

A A-2, B-1, C-4, D-3 (or A-2, B-4, C-1, D-3)	B A-1, B-2, C-4, D-3
C A-2, B-3, C-1, D-4	D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Adhai Din Ka Jhonpra (Ajmer):** Built by **Qutbuddin Aibak** in 1199 on the ruins of a Sanskrit college (*Saraswati Mandir*) built by Chauhan king Vigraharaja IV (Visaladeva).
- **Alai Darwaza (Qutb complex, 1311):** Masterpiece of **Alauddin Khalji**; first building in India to employ true horseshoe arch and dome on squinches.
- **Siri Fort & Hauz Khas (Hauz-i-Alai):** Built by **Alauddin Khalji** as his military capital to defend Delhi from Mongol onslaughts.
- **Tughlaqabad Fort:** Massive fortified stone city built by **Ghiyasuddin Tughlaq** (Ghazi Malik) in 1321, characterized by sloping battered walls (*batter*).

⚠️ आयोग का TRAP / RATTA POINT

Qutb Minar construction was initiated by **Qutbuddin Aibak** (1st storey), completed by **Iltutmish** (added 3 storeys), and repaired by **Firoz Shah Tughlaq** (replaced damaged 4th storey with two storeys of marble and sandstone)!

Q. 193 Indian Economy | 15th Finance Commission Tax Devolution Criteria

Match List-I (Devolution Criterion of 15th Finance Commission) with List-II (Assigned Percentage Weight) and select the correct answer:

List-I (Criterion)	List-II (Weight %)
A Income Distance	1 15.0%
B Population (Census 2011)	2 45.0%
C Forest and Ecology	3 12.5%
D Demographic Performance	4 10.0%

A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- The 15th Finance Commission (chaired by N.K. Singh) recommended retaining vertical devolution of central divisible tax pool to States at **41%** (adjusted from 42% due to creation of UTs of J&K and Ladakh).
- **Horizontal Devolution Weights (Total 100%):**
    1. **Income Distance: 45.0%** (largest weight; distance from highest per capita income state Goa/Haryana).
    2. **Population (Census 2011 exclusively): 15.0%**.
    3. **Area: 15.0%**.
    4. **Forest and Ecology: 10.0%** (dense forest cover share).
    5. **Demographic Performance: 12.5%** (rewards states that successfully controlled Total Fertility Rate - TFR).
    6. **Tax and Fiscal Efforts: 2.5%**.

⚠ आयोग का **TRAP / RATTA POINT**

15th FC used **Census 2011 data exclusively**, completely discarding the Census 1971 population data used by previous commissions!

Q. 194 Current Affairs | Global Hunger Index (GHI) 2024

Consider the following statements regarding the Global Hunger Index (GHI) 2024:

1. GHI is jointly published annually by Welthungerhilfe (Germany) and Concern Worldwide (Ireland).
2. The index scores hunger based on four indicators: Undernourishment, Child Stunting, Child Wasting, and Child Mortality.
3. In GHI 2024, India was placed in the "Serious" hunger category, ranking 105th out of 127 countries.

Which of the statements given above are correct?

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

✓ Correct Answer :Option D

DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
    - **4 Indicators across 3 Dimensions:**
    - *Inadequate food supply:* Undernourishment (share of population with insufficient caloric intake, weight $\frac{1}{3}$).
    - *Child undernutrition:* Child Wasting (low weight-for-height under age 5, acute undernutrition, weight $\frac{1}{6}$) and Child Stunting (low height-for-age under age 5, chronic undernutrition, weight $\frac{1}{6}$).
    - *Child mortality:* Under-five mortality rate (weight $\frac{1}{3}$).
    - **India in GHI 2024:** Ranked **105th out of 127 countries** with a score of 27.3, categorizing its hunger level as "Serious". India recorded the world's highest child wasting rate (~18.7%).

⚠ आयोग का **TRAP / RATTA POINT**

In GHI scoring: **0 is the best score (No hunger)** and **100 is the worst score**!

Q. 195 📜 Modern Indian History 📌 Peasant Associations in Uttar Pradesh

Consider the following peasant organizations formed in colonial Uttar Pradesh:

1. The **UP Kisan Sabha** was set up in February 1918 by Gauri Shankar Mishra, Indra Narayan Dwivedi, and supported by Madan Mohan Malaviya.
2. The **Awadh Kisan Sabha** was founded in October 1920 at Pratapgarh, actively mobilized by Baba Ramchandra.
3. The **Eka Movement** (1921–1922) spread across Hardoi, Bahraich, and Sitapur districts, led by Madari Pasi.

Which of the statements given above are correct?

A 1 and 2 only	B 2 and 3 only
C 1 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **UP Kisan Sabha (Feb 1918):** Formed in Allahabad by Gauri Shankar Mishra and Indra Narayan Dwivedi with Madan Mohan Malaviya's backing; established over 450 village branches.
  - **Awadh Kisan Sabha (Oct 1920):** Formed in Pratapgarh due to ideological divergence with Malaviya's moderate approach; led by **Baba Ramchandra** (a Maharashtrian who recited Tulsidas's *Ramcharitmanas* to mobilize peasants) and Jawaharlal Nehru.
  - **Eka Movement (Unity Movement, late 1921):** Grassroots tenant movement in northern Awadh (Hardoi, Bahraich, Sitapur); spearheaded by low-caste leader **Madari Pasi**; resisted 50% above-recorded rents and forced labor (*Begar*).

⚠️ आयोग का TRAP / RATTA POINT

All India Kisan Sabha (AIKS) was founded at Lucknow in **April 1936** with **Swami Sahajanand Saraswati** as President and **N.G. Ranga** as General Secretary!

Q. 196 🌍 Geography of India 📌 Longitudinal (Regional) Divisions of the Himalayas

Match List-I (Regional Himalayan Division) with List-II (Demarcating Rivers / Extent) and select the correct answer:

List-I (Division)	List-II (River Extent)
A Kashmir / Punjab Himalaya	1 Between Sutlej and Kali rivers
B Kumaon Himalaya	2 Between Indus and Sutlej rivers
C Nepal Himalaya	3 Between Teesta and Dihang (Brahmaputra) rivers
D Assam Himalaya	4 Between Kali and Teesta rivers

A A-2, B-1, C-4, D-3	B A-1, B-2, C-4, D-3
C A-2, B-4, C-1, D-3	D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- Proposed by Sir Sidney Burrard based on river valleys:
  - **Kashmir / Punjab Himalaya:** Between **Indus and Sutlej** (~560 km); contains Karakoram, Ladakh, Pir Panjal, Zoji La; Karewa formations for saffron cultivation.
  - **Kumaon Himalaya:** Between **Sutlej and Kali** (~320 km) in Uttarakhand; contains Nanda Devi, Kamet, Badrinath, Kedarnath, Trishul peaks.
  - **Nepal Himalaya:** Between **Kali and Teesta** (~800 km); longest section; contains tallest peaks: Mount Everest, Kanchenjunga, Makalu, Dhaulagiri, Annapurna.
  - **Assam Himalaya:** Between **Teesta and Dihang** (~720 km); contains Namcha Barwa, Kula Kangri.

⚠️ आयोग का TRAP / RATTA POINT

Longest regional division = **Nepal Himalaya (800 km)**; Highest peak in Kumaon Himalaya = **Nanda Devi (7,817 m)**!

Q. 197 🏛️ Indian Polity 📌 Comptroller and Auditor General (CAG) Duties

Consider the following statements regarding the Comptroller and Auditor General of India (CAG):

1. Under Article 149, the CAG audits all expenditure from the Consolidated Fund of India, Consolidated Fund of each State, and Consolidated Fund of each Union Territory having a Legislative Assembly.
2. The CAG submits audit reports relating to the accounts of the Union to the President, who causes them to be laid before each House of Parliament under Article 151.
3. The CAG holds office during the pleasure of the President of India.

Which of the statements given above is/are correct?

A 1 and 2 only	**B** 2 and 3 only
C 1 only	**D** 1, 2 and 3

✓ **Correct Answer :Option A**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- Statements 1 and 2 are correct: The CAG's duties (under CAG DPC Act 1971) encompass auditing all receipts and expenditures of Consolidated Funds, Contingency Funds, and Public Accounts of Centre and States. Under Article 151(1), Union reports go to the President (laid before Parliament); under Article 151(2), State reports go to the Governor (laid before State Legislature).
- Statement 3 is incorrect: The CAG does **NOT hold office during the pleasure of the President**! The CAG enjoys constitutional security of tenure (6 years or until age 65) and can be removed ONLY in like manner and on like grounds as a Supreme Court judge (impeachment by special majority of Parliament on grounds of proved misbehavior or incapacity).

⚠️ आयोग का TRAP / RATTA POINT

Post-retirement restriction: The CAG is **ineligible for further office** under the Government of India or any State Government after retirement (Article 148(4))!

Q. 198 🔬 General Science (Biology) 📌 Photosynthesis & Chloroplast Biochemistry

Consider the following statements regarding photosynthesis:

1. Chlorophyll pigment contains an atom of Magnesium (Mg) coordinated at the center of its porphyrin ring.
2. The Light-dependent Reaction (photochemical phase) takes place in the thylakoid membranes / grana of the chloroplast, producing ATP and $NADPH$.
3. The Dark Reaction (Calvin cycle / biosynthetic phase) occurs in the stroma of the chloroplast and does not require light directly.

Which of the statements given above are correct?

A 1 and 2 only	**B** 2 and 3 only
C 1 and 3 only	**D** 1, 2 and 3

✓ **Correct Answer :Option D**

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Chlorophyll:** Green photosynthetic pigment; contains a **Magnesium (Mg^{2+})** ion in a tetrapyrrole porphyrin ring (similar to Iron in hemoglobin).
  - **Light Reaction:** Photolysis of water ($2H_2O \rightarrow 4H^+ + 4e^- + O_2$) occurs on the inner side of thylakoid membrane in PS-II; oxygen gas is released; ATP and NADPH are synthesized (Assimilatory Power).
  - **Dark Reaction (Calvin Cycle):** Occurs in the **Stroma**; catalyzed by the enzyme **RuBisCO** (most abundant enzyme on Earth); fixes atmospheric CO_2 into glucose using ATP and NADPH.

⚠️ आयोग का TRAP / RATTA POINT

The Oxygen (O_2) evolved during photosynthesis comes from **WATER (H_2O)**, NOT from Carbon Dioxide (CO_2)! (Proven by Cornelius van Niel and Ruben & Kamen using oxygen isotope ^{18}O).

Q. 199 🎯 Uttar Pradesh Special 📌 Classical Music Gharanas of Uttar Pradesh

Match List-I (Music Gharana of UP) with List-II (Legendary Exponent) and select the correct answer:

List-I (Gharana)	List-II (Master Exponent)
A Lucknow Gharana	1 Ustad Bismillah Khan (Shehnai) & Pandit Kishan Maharaj (Tabla)
B Benaras Gharana	2 Pandit Birju Maharaj & Lachhu Maharaj (Kathak Dance)
C Kirana Gharana	3 Ustad Faiyaz Khan (*Aftab-e-Mousiqi*)
D Agra Gharana	4 Pandit Bhimsen Joshi & Gangubai Hangal (Khayal Vocals)

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Lucknow Gharana:** Developed under Nawab Wajid Ali Shah (pen name *Akhtarpiya*); legendary Kathak exponents: Kalka Prasad, Bindadin Maharaj, Achchan Maharaj, Lachhu Maharaj, **Pandit Birju Maharaj**; also famous for Khayal, Thumri, and Tabla (Modu Khan, Bakshu Khan).
- **Benaras Gharana:** Famous for Shehnai (**Ustad Bismillah Khan**, Bharat Ratna 2001), Tabla (Pandit Kishan Maharaj, Samta Prasad "Gudai Maharaj"), and Thumri (Girija Devi, Siddheshwari Devi).
- **Kirana Gharana:** Founded by Ustad Abdul Karim Khan and Abdul Wahid Khan (named after Kairana in Shamli/Muzaffarnagar, UP); celebrated by **Pandit Bhimsen Joshi** (Bharat Ratna 2008) and Gangubai Hangal.
- **Agra Gharana (Rangila Gharana):** Associated with Haji Sujan Khan and perfected by **Ustad Faiyaz Khan**.

⚠️ आयोग का TRAP / RATTA POINT

Ustad Bismillah Khan was awarded the Bharat Ratna in **2001** (along with Lata Mangeshkar)!

Q. 200 🌿 Environment & Ecology 📌 Bioaccumulation and Biomagnification

Consider the following statements:

1. 'Bioaccumulation' refers to the gradual accumulation of a toxic chemical in the tissues of a single organism over its lifetime.
2. 'Biomagnification' refers to the progressive increase in the concentration of a persistent non-biodegradable pollutant at successive higher trophic levels in a food chain.
3. For biomagnification to occur, a pollutant must be long-lived, mobile, soluble in fats (lipophilic), and biologically active.

Which of the statements given above are correct?

A 1 and 2 only

B 2 and 3 only

C 1 and 3 only

D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Bioaccumulation:** Takes place within a **single trophic level** (intake rate exceeds metabolic excretion/breakdown rate).
  - **Biomagnification:** Occurs **across multiple trophic levels** (concentration multiplies up the food chain).
  - Classic examples:
  - **DDT:** Concentration in water (0.003 ppb) biomagnifies through zooplankton (0.04 ppm), small fish (0.5 ppm), fish-eating birds (25 ppm) by millions of times, causing eggshell thinning and bird population collapse.
  - **Methylmercury:** Concentrates in predatory fish (tuna, shark), causing neurological damage (Minamata disease).

⚠️ आयोग का TRAP / RATTA POINT

Water-soluble toxins do NOT biomagnify because they are rapidly excreted in urine. ONLY **fat-soluble (lipophilic)**, persistent pollutants biomagnify!

Q. 201 🌍 World Geography 📌 Major Rivers of Europe & North America

Match List-I (River) with List-II (Distinct Feature / Outflow) and select the correct answer:

List-I (River)	List-II (Feature / Outflow)
A Danube River	1 Empties into Caspian Sea; longest river in Europe
B Rhine River	2 Flows through or touches 10 countries and passes through 4 national capitals (Vienna, Bratislava, Budapest, Belgrade)
C St. Lawrence River	3 Connects the Great Lakes with the Atlantic Ocean; busiest inland waterway in North America
D Volga River	4 Flows through the Ruhr industrial valley into the North Sea at Rotterdam

A A-2, B-4, C-3, D-1

B A-1, B-2, C-4, D-3

C A-2, B-3, C-4, D-1

D A-3, B-4, C-1, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Danube:** Originates in Black Forest (Germany); flows eastward for 2,850 km into the Black Sea; passes through 4 European capitals: **Vienna (Austria), Bratislava (Slovakia), Budapest (Hungary), and Belgrade (Serbia)**.
- **Rhine:** Originates in Swiss Alps; flows through Germany's industrial heartland (Ruhr) into the North Sea at Rotterdam (Netherlands).
- **St. Lawrence:** Drains the entire Great Lakes basin into the Atlantic Ocean (Gulf of St. Lawrence); forms part of the international boundary between USA and Canada.
- **Volga:** Longest river in Europe (~3,530 km); flows entirely within Russia and drains into the enclosed **Caspian Sea**.

⚠️ आयोग का TRAP / RATTA POINT

Longest river in Europe is the **Volga**; second longest is the **Danube**!

Q. 202 📈 Indian Economy 📌 Flagship Agricultural Welfare Schemes

Consider the following statements regarding the Pradhan Mantri Fasal Bima Yojana (PMFBY):

1. It was launched in 2016, replacing the National Agricultural Insurance Scheme (NAIS).
2. The uniform premium payable by farmers is 2% for all Kharif food and oilseeds crops.
3. The uniform premium payable by farmers is 1.5% for all Rabi food and oilseeds crops, and 5% for commercial/horticultural crops.

Which of the statements given above are correct?

A 1 and 2 only

B 2 and 3 only

C 1 and 3 only

D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **PMFBY Launch:** Introduced in Kharif 2016 by Ministry of Agriculture & Farmers Welfare.
  - **Uniform Farmer Premium Caps:**
  - **Kharif Crops (Food & Oilseeds): 2.0%** of sum insured.
  - **Rabi Crops (Food & Oilseeds): 1.5%** of sum insured.
  - **Annual Commercial / Horticultural Crops: 5.0%** of sum insured.
  - The balance premium is subsidized equally (50:50) by the Central and State Governments (90:10 for Northeastern states).

⚠️ आयोग का TRAP / RATTA POINT

Farmer premium rates are fixed: **2% Kharif, 1.5% Rabi, 5% Horticultural**!

Q. 203 Current Affairs | Nobel Prize in Chemistry 2024

Consider the following statements regarding the Nobel Prize in Chemistry 2024:

1. One half of the prize was awarded to David Baker for "computational protein design".
2. The other half was awarded jointly to Demis Hassabis and John M. Jumper for "protein structure prediction" using the AI model AlphaFold2.
3. AlphaFold2 was developed by Google DeepMind.

Which of the statements given above are correct?

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - The Royal Swedish Academy of Sciences awarded the 2024 Nobel Prize in Chemistry to three scientists:
  - **David Baker (University of Washington):** Succeeded in the almost impossible feat of building entirely new kinds of proteins using his software *Rosetta*.
  - **Demis Hassabis and John M. Jumper (Google DeepMind):** Developed the revolutionary AI system **AlphaFold2**, solving a 50-year-old biological grand challenge by predicting the complex 3D structures of virtually all 200 million known proteins from amino acid sequences.

⚠️ आयोग का **TRAP / RATTA POINT**

Note that AlphaFold predicts **3D Protein Structures** from 1D amino acid sequences!

Q. 204 Ancient Indian History | Ashokan Pillars and Animal Capitals

Match List-I (Ashokan Pillar Site) with List-II (Animal Capital Surmounting It) and select the correct answer:

List-I (Site)	List-II (Animal Capital)
A Sarnath Pillar	1 Bull Capital (now in Rashtrapati Bhavan)
B Rampurva Pillar (Bihar)	2 Four Addorsed Lions (National Emblem of India)
C Sankisa Pillar (Uttar Pradesh)	3 Single Lion Capital
D Lauriya Nandangarh (Bihar)	4 Elephant Capital

A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Sarnath Pillar:** Four Asiatic lions seated back-to-back on an abacus depicting Elephant, Horse, Bull, and Lion separated by 24-spoke Dharma wheels; adopted as India's State Emblem on 26 January 1950.
- **Rampurva (West Champaran, Bihar):** Two pillars discovered: one crowned by a **Bull** (now preserved at the entrance of Rashtrapati Bhavan, New Delhi) and the other by a Lion.
- **Sankisa (Farrukhabad, UP):** Carries a carved **Elephant** capital.
- **Lauriya Nandangarh (Bihar):** Beautifully preserved monolithic polished sandstone pillar surmounted by a **Single Lion** facing east.

⚠️ आयोग का **TRAP / RATTA POINT**

Rampurva Bull capital is installed at Rashtrapati Bhavan. Basarah-Bakhira (Vaishali) pillar has a single seated Lion facing north!

Q. 205 🏛️ Indian Polity 📌 Constitutional Powers of the State Governor

Consider the following statements regarding the Governor of a State:

1. Under Article 153, there shall be a Governor for each State, but the 7th Constitutional Amendment Act, 1956 permitted the appointment of the same person as Governor for two or more States.
2. Under Article 213, the Governor can promulgate an ordinance when the State Legislative Assembly is not in session.
3. Under Article 200, when a Bill is passed by the State Legislature, the Governor may reserve the bill for the consideration of the President, and reservation is mandatory if the bill endangers the position of the High Court.

Which of the statements given above are correct?

A 1 and 2 only
B 2 and 3 only
C 1 and 3 only
D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Article 153:** 7th CAA 1956 added the proviso that the same person may be appointed Governor of two or more states.
  - **Article 213 (Ordinance Power):** Governor can issue ordinances only when assembly (or either house in bicameral states) is not in session; must be approved by the legislature within **6 weeks** of its reassembly.
  - **Article 200 (Reservation for President):** Discretionary power to reserve bills; second proviso makes it **obligatory** to reserve any bill that derogates from the constitutional powers of the High Court. Under Article 201, the President may give assent, withhold assent, or direct return of the bill.

⚠️ आयोग का TRAP / RATTA POINT

Unlike the Governor, the President has NO time limit to decide on a reserved state bill, and the President is NOT bound to give assent even if the State Legislature repasses the bill!

Q. 206 📜 Medieval Indian History 📌 Bhakti Movement Saints Chronology

Arrange the following Bhakti saints in ascending chronological order of their life periods:

1. Ramanujacharya
2. Kabir Das
3. Guru Nanak Dev
4. Chaitanya Mahaprabhu

Select the correct answer from the codes given below:

A 1 — 2 — 3 — 4
B 2 — 1 — 3 — 4
C 1 — 3 — 2 — 4
D 1 — 2 — 4 — 3

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

1. **Ramanujacharya (1017–1137 CE):** South Indian philosopher-saint; propounded *Vishishtadvaita* Vedanta; revived Sri Vaishnavism.
2. **Kabir Das (c. 1398–1448 / 1518 CE):** Nirguna Bhakti poet-weaver of Varanasi; disciple of Ramananda; composed *Dohas* collected in the *Bijak* (Sakhi, Sabad, Ramaini).
3. **Guru Nanak Dev (1469–1539 CE):** Founder of Sikhism; born at Talwandi (Nankana Sahib); established *Langar* and *Sangat*.
4. **Chaitanya Mahaprabhu (1486–1534 CE):** Mystic saint of Bengal; popularized *Kirtan* (congregational chanting of Hare Krishna mantra); Gaudiya Vaishnavism.

⚠️ आयोग का TRAP / RATTA POINT

Chronology sequence: **Shankaracharya (8th C) -> Ramanuja (11th C) -> Ramananda (14th C) -> Kabir (15th C) -> Guru Nanak (1469) -> Chaitanya (1486) -> Mirabai (1498) -> Tulsidas (1532)**!

Q. 207 🔬 General Science (Physics) 📌 Surface Tension and Capillarity

Consider the following everyday phenomena:

1. Raindrops and small liquid droplets assume a spherical shape.
2. Oil rises up through the cotton wick of an earthen lamp.
3. Adding soap or detergent to water helps remove grease and dirt from clothes.

Which of the above phenomena are directly governed by the physical property of **Surface Tension**?

A 1 and 2 only	B 1 and 3 only
C 2 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three phenomena are governed by surface tension:
  - **Spherical Droplets:** Liquid surfaces tend to minimize surface energy and contract to the minimum possible surface area for a given volume, which is a **Sphere**.
  - **Lamp Wick (Capillarity):** Driven by surface tension and adhesive forces between liquid molecules and narrow pores of the fabric ($h = \frac{2T \cos\theta}{r\rho g}$).
  - **Detergents & Soaps:** Surfactants significantly **lower the surface tension** of water, increasing its wetting power so it can penetrate microscopic pores in fabrics and emulsify grease.

⚠️ आयोग का TRAP / RATTA POINT

As temperature increases, surface tension of liquids **DECREASES** (and becomes zero at the critical temperature)!

Q. 208 🎯 Uttar Pradesh Special 📌 Traditional Fairs & Locations (Part 2)

Match List-I (Fair / Mela in UP) with List-II (District / Location) and select the correct answer:

List-I (Fair)	List-II (District / Location)
A Lathmar Holi	1 Kasganj
B Soron Fair	2 Barsana and Nandgaon (Mathura)
C Govind Saheb Fair	3 Balrampur
D Devi Patan Fair	4 Ambedkar Nagar

A A-2, B-1, C-4, D-3	B A-1, B-2, C-4, D-3
C A-2, B-4, C-1, D-3	D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Lathmar Holi:** Celebrated days before Holi in **Barsana** (Radha's birthplace) and **Nandgaon** (Krishna's village) in Mathura district, where women playfully beat men with wooden sticks (*lathis*).
- **Soron Fair:** Held at Soron (Sukar Kshetra) on the banks of the Ganges in **Kasganj** district (traditional birthplace of saint-poet Goswami Tulsidas).
- **Govind Saheb Fair:** Famous month-long cattle and rural fair held at Jalalpur in **Ambedkar Nagar** district (honoring saint Govind Saheb).
- **Devi Patan Fair:** Held at the Pateshwari Devi temple (one of the 51 Shaktipeethas) in **Balrampur** district during Chaitra Navratri.

⚠️ आयोग का TRAP / RATTA POINT

Maximum number of fairs in Uttar Pradesh are held in **Mathura district** (approx. 86 fairs annually)!

Q. 209 Environment & Ecology | Project Rhino and Greater One-Horned Rhinoceros

Consider the following statements regarding the Greater One-Horned Rhinoceros (*Rhinoceros unicornis*):

1. India is home to over 80% of the global population of the Greater One-Horned Rhinoceros.
2. Kaziranga National Park in Assam holds the highest single population of wild rhinos in the world.
3. The species is classified as 'Critically Endangered' on the IUCN Red List.

Which of the statements given above is/are correct?

A. 1 and 2 only
B. 2 and 3 only
C. 1 only
D. 1, 2 and 3

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- Statements 1 and 2 are correct: Global population exceeds 4,000 rhinos, with ~3,200 residing in India (primarily Assam, West Bengal, and UP). Kaziranga National Park alone hosts over 2,600 rhinos. In UP, a successful rhino reintroduction program has been running in **Dudhwa National Park** since 1984.
- Statement 3 is incorrect: The Greater One-Horned Rhino is classified as **'Vulnerable'** (NOT Critically Endangered) on the IUCN Red List, having been downlisted due to successful anti-poaching and conservation efforts under Indian Rhino Vision 2020 (IRV 2020).

⚠️ आयोग का TRAP / RATTA POINT

Javan Rhino and Sumatran Rhino are Critically Endangered, but the Indian Greater One-Horned Rhino is **Vulnerable**! Protected under Schedule I of Wildlife Protection Act 1972.

Q. 210 Indian Economy | Regulatory Institutions of India

Match List-I (Regulatory Institution) with List-II (Statutory Act / Headquarters) and select the correct answer:

List-I (Institution)	List-II (Detail / Location)
A Securities and Exchange Board of India (SEBI)	1 Established 1982 on B. Sivaraman Committee report; apex rural development bank
B Insurance Regulatory and Development Authority of India (IRDAI)	2 Statutory status granted in 1992 under SEBI Act; capital markets regulator in Mumbai
C National Bank for Agriculture and Rural Development (NABARD)	3 Statutory body under IRDA Act 1999; headquarters located at Hyderabad
D Pension Fund Regulatory and Development Authority (PFRDA)	4 Regulates National Pension System (NPS); headquartered in New Delhi

A. A-2, B-3, C-1, D-4
B. A-1, B-2, C-4, D-3
C. A-2, B-1, C-4, D-3
D. A-3, B-2, C-1, D-4

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **SEBI:** Non-statutory body created in 1988; given full statutory autonomous regulatory powers via SEBI Act 1992 (headquarters in Mumbai).
- **IRDAI:** Created on recommendations of R.N. Malhotra Committee (1994); statutory body under IRDA Act 1999; relocated its headquarters from Delhi to **Hyderabad** in 2001.
- **NABARD:** Formed on July 12, 1982, under Act 61 of 1981 based on the B. Sivaraman Committee recommendations; dedicated to agricultural credit and rural development.
- **PFRDA:** Established in 2003, statutory status under PFRDA Act 2013; regulates pensions and NPS.

⚠️ आयोग का TRAP / RATTA POINT

IRDAI headquarters is in **Hyderabad**, NOT Mumbai or New Delhi! SIDBI headquarters is in **Lucknow**!

Q. 211 Current Affairs | Nobel Prize in Physics 2024

The Nobel Prize in Physics 2024 was awarded jointly to John J. Hopfield and Geoffrey E. Hinton for which breakthrough achievement?

A Discoveries in topological phase transitions and topological phases of matter

B Foundational discoveries and inventions that enable machine learning with artificial neural networks

C Groundbreaking contributions to the understanding of complex physical systems and climate modeling

D Theoretical discoveries of topological defects in quantum materials

✓ Correct Answer :Option B

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- The Royal Swedish Academy of Sciences awarded the 2024 Nobel Prize in Physics to **John J. Hopfield (Princeton University)** and **Geoffrey E. Hinton (University of Toronto - "Godfather of AI")**.
- Recognized "for foundational discoveries and inventions that enable machine learning with artificial neural networks":
  - *Hopfield:* Invented the **Hopfield Network**, an associative memory that can store and reconstruct images and data patterns using physical principles of atomic spin systems.
  - *Hinton:* Developed the **Boltzmann Machine**, which autonomously discovers properties in data and laid the groundwork for modern Deep Learning.

⚠️ आयोग का TRAP / RATTA POINT

This marked the first time a Nobel Prize in Physics was awarded for core computer science / artificial intelligence algorithms!

Q. 212 Modern Indian History | British Colonial Acts (1909, 1919, 1935 Comparison)

Match List-I (British Enactment) with List-II (Key Constitutional Innovation) and select the correct answer:

List-I (Act)	List-II (Innovation)
A Indian Councils Act 1909 (Morley-Minto Reforms)	1 Introduced Provincial Autonomy, abolished Diarchy in Provinces, and proposed an All-India Federation
B Government of India Act 1919 (Montagu-Chelmsford Reforms)	2 Introduced Communal Electorates (Separate Electorates for Muslims) for the first time
C Government of India Act 1935	3 Introduced Diarchy in the Provincial Executive (Transferred and Reserved Subjects) and Bicameralism at the Centre

A A-2, B-3, C-1

B A-1, B-3, C-2

C A-2, B-1, C-3

D A-3, B-2, C-1

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Act of 1909 (Morley-Minto):** Legalized communalism; Lord Minto was hailed as "Father of Communal Electorate"; allowed Indians into Viceroy's Executive Council (Satyendra Prasad Sinha was the first Indian member as Law Member).
- **Act of 1919 (Mont-Ford):** Introduced **Diarchy in 8 Provinces** (Reserved subjects like Law & Order/Finance under British Governor; Transferred subjects like Education/Health under Indian Ministers); introduced direct elections and bicameralism at Centre.
- **Act of 1935:** Abolished provincial diarchy; granted **Provincial Autonomy**; introduced Diarchy at the Centre (never implemented); created Federal Court (1937) and Reserve Bank of India.

⚠️ आयोग का TRAP / RATTA POINT

Diarchy in Provinces = **1919 Act**; Diarchy at Centre & Provincial Autonomy = **1935 Act**!

Q. 213 🌍 World Geography 📌 World Famous Mineral Belts & Mines

Match List-I (Famous World Mining Center) with List-II (Principal Mineral Mined) and select the correct answer:

List-I (Mining Center)	List-II (Mineral)
A Witwatersrand (South Africa)	1 Diamond
B Kimberley (South Africa)	2 Gold
C Sudbury Basin (Ontario, Canada)	3 Copper
D Chuquicamata (Chile)	4 Nickel and Platinum Group Metals

A. A-2, B-1, C-4, D-3
B. A-1, B-2, C-4, D-3
C. A-2, B-4, C-1, D-3
D. A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 **DETAILED SOLUTION & HIGH-YIELD ANALYSIS**

- **Witwatersrand ("The Rand", South Africa):** Greatest gold-producing reef in the world; produced over 40% of all gold ever mined on Earth.
- **Kimberley (South Africa):** Famous for the "Big Hole"; historic capital of diamond mining (Kimberlite rock pipes).
- **Sudbury Basin (Canada):** Ancient meteorite impact crater; world's leading producer of Nickel and copper.
- **Chuquicamata (Atacama Desert, Chile):** Largest open-pit copper mine in the world by excavated volume.

⚠️ आयोग का **TRAP / RATTA POINT**

Katanga (DR Congo) / Copperbelt (Zambia) = **Copper and Cobalt**; Broken Hill (Australia) = **Lead, Zinc, Silver**!

Q. 214 🏛️ Indian Polity 📌 Union Public Service Commission (UPSC)

Consider the following statements regarding the Union Public Service Commission (UPSC):

1. UPSC is established as an independent constitutional body under Articles 315 to 323 in Part XIV of the Constitution.
2. The Chairman and members of UPSC hold office for a term of six years or until they attain the age of 65 years, whichever is earlier.
3. The Chairman of UPSC can be removed from office by the President on the ground of misbehavior only after an inquiry by the Supreme Court under Article 317, and the advice of the Supreme Court is binding on the President.

Which of the statements given above are correct?

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

✓ Correct Answer :Option D

📘 **DETAILED SOLUTION & HIGH-YIELD ANALYSIS**

- All three statements are correct:
  - **Constitutional Mandate:** Article 315 provides for a Public Service Commission for the Union and for each State.
  - **Tenure (Article 316):** 6 years or age 65 for UPSC (for State PSCs, it is 6 years or age 62).
  - **Removal (Article 317):** President refers matter of alleged misbehavior to Supreme Court. SC conducts inquiry; if SC recommends removal, **the President is constitutionally BOUND to remove the Chairman/member**.

⚠️ आयोग का **TRAP / RATTA POINT**

Post-retirement: Chairman of UPSC is **completely ineligible for any further employment** under Central or State Government!

Q. 215 🔬 General Science (Biology) 📌 Human Eye Anatomy & Photoreceptors

Consider the following statements regarding the human eye:

1. The Cornea is the transparent front part of the eye that lacks blood vessels and can be transplanted without immunological rejection.
2. The Retina contains two types of photoreceptor cells: Rods (responsible for twilight/scotopic vision) and Cones (responsible for daylight/photopic and color vision).
3. The light-sensitive pigment in Rod cells is Rhodopsin (visual purple), which requires Vitamin A for its synthesis.

Which of the statements given above are correct?

A 1 and 2 only
B 2 and 3 only
C 1 and 3 only
D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Cornea:** Avascular (receives oxygen directly from atmospheric air and nutrients from aqueous humor); high success rate in corneal grafts (*Keratoplasty*).
  - **Rods vs Cones:**
  - *Rods (~120 million):* Contain **Rhodopsin**; sensitive to low light (night vision); cannot distinguish color. Vitamin A deficiency impairs rhodopsin synthesis, causing **Night Blindness (Nyctalopia)**.
  - *Cones (~6 million):* Concentrated at the *Fovea centralis* (yellow spot); contain photopsins; provide high-resolution daylight vision and perceive red, green, and blue colors.

⚠️ आयोग का TRAP / RATTA POINT

At the **Blind Spot (Optic Disc)**, there are NO rods or cones, hence no image can be formed there!

Q. 216 🎯 Uttar Pradesh Special 📌 Wildlife Sanctuaries of UP (Superlatives)

Match List-I (Wildlife Sanctuary in UP) with List-II (Distinct Feature / Location) and select the correct answer:

List-I (Sanctuary)	List-II (Feature / District)
A Hastinapur Wildlife Sanctuary	1 Smallest wildlife sanctuary in UP (5.4 sq km); located in Lalitpur district
B Mahavir Swami Wildlife Sanctuary	2 Largest wildlife sanctuary in UP (2,073 sq km); spans Meerut, Muzaffarnagar, Bijnor, Hapur, Amroha
C National Chambal Wildlife Sanctuary	3 Bahraich district (Terai belt); vital eco-corridor between Dudhwa and Bardia (Nepal)
D Katerniaghat Wildlife Sanctuary	4 Tri-state protected area (UP, MP, Rajasthan); famous for critically endangered Gharial and Gangetic Dolphin

A A-2, B-1, C-4, D-3
B A-1, B-2, C-4, D-3
C A-2, B-4, C-1, D-3
D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Hastinapur WLS: Largest** wildlife sanctuary in Uttar Pradesh (~2,073 sq km); established in 1986; home to Swamp Deer (*Barasingha*).
- **Mahavir Swami WLS: Smallest** wildlife sanctuary in UP (~5.4 sq km); located in **Lalitpur** district.
- **National Chambal Sanctuary:** Riverine sanctuary on Chambal river spanning UP (Agra/Etawah), MP, and Rajasthan; premier sanctuary for Gharials (*Gavialis gangeticus*) and Red-crowned roofed turtles.
- **Katerniaghat WLS:** Located in **Bahraich** district along the Girwa river; connects Dudhwa Tiger Reserve with Nepal's Bardia National Park.

⚠️ आयोग का TRAP / RATTA POINT

Chandraprabha Wildlife Sanctuary (Chandauli, established 1957) was the **FIRST wildlife sanctuary** established in Uttar Pradesh!

Q. 217 🌿 Environment & Ecology 📌 Ecosystem Productivity (GPP vs NPP)

Consider the following statements regarding ecological productivity:

1. Gross Primary Productivity (GPP) is the total rate of production of organic matter during photosynthesis.
2. Net Primary Productivity (NPP) is the remaining biomass available to heterotrophs after accounting for respiration losses (R), represented as $\text{NPP} = \text{GPP} - R$.
3. Among global ecosystems, open oceans have the highest net primary productivity per unit area, while tropical rainforests have the lowest.

Which of the statements given above is/are correct?

A 1 and 2 only	**B** 2 and 3 only
C 1 only	**D** 1, 2 and 3

✓ **Correct Answer :Option A**

📘 **DETAILED SOLUTION & HIGH-YIELD ANALYSIS**

- Statements 1 and 2 are correct: GPP measures total solar energy fixed into organic chemical bonds. A significant portion of this energy is consumed by plants for their own cellular respiration (R). What remains as standing biomass is $\text{NPP} = \text{GPP} - R$.
- Statement 3 is incorrect: The situation is the OPPOSITE!
  - **Highest productivity per unit area ($g/m^2/year$): Tropical Rainforests, Estuaries, and Coral Reefs** (over $2,000 \text{ g/m}^2/\text{yr}$).
  - **Lowest productivity per unit area: Extreme Deserts, Tundra, and Deep Open Oceans** (less than $125 \text{ g/m}^2/\text{yr}$).
  - (However, because Open Oceans cover ~71% of Earth's surface, their *total collective global productivity* is large).

⚠️ **आयोग का TRAP / RATTA POINT**

Coral reefs and tropical rainforests have the HIGHEST productivity per unit area; open oceans have LOW productivity per unit area!

Q. 218 📈 Indian Economy 📌 Components of Foreign Exchange Reserves of India

India's Foreign Exchange Reserves managed by the Reserve Bank of India comprise how many distinct components?

A Two	**B** Three
C Four	**D** Five

✓ **Correct Answer :Option C**

📘 **DETAILED SOLUTION & HIGH-YIELD ANALYSIS**

- India's forex reserves consist of **4 distinct components**:
  1. **Foreign Currency Assets (FCA):** Largest component (~85–88% of total); held in multi-currency assets (US Dollars, Euros, Pound Sterling, Japanese Yen).
  2. **Gold Reserves:** Physical gold held by RBI (stored in RBI vaults and Bank of England).
  3. **Special Drawing Rights (SDR):** International reserve asset created by the IMF in 1969.
  4. **Reserve Tranche Position (RTP):** The reserve tranche that India maintains with the International Monetary Fund (IMF).

⚠️ **आयोग का TRAP / RATTA POINT**

Foreign Currency Assets (FCA) is the LARGEST component; Gold is the second largest!

Q. 219 Current Affairs | Nobel Prize in Literature 2024

Who was awarded the Nobel Prize in Literature 2024 "for her intense poetic prose that confronts historical traumas and exposes the fragility of human life"?

- **A** Han Kang (South Korea)
- **B** Jon Fosse (Norway)
- **C** Annie Ernaux (France)
- **D** Can Xue (China)

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- South Korean author **Han Kang** was awarded the 2024 Nobel Prize in Literature.
- She became the **first South Korean writer** and the **first Asian woman** to win the Nobel Prize in Literature.
- Famous works: *The Vegetarian* (won International Booker Prize 2016), *Human Acts*, and *We Do Not Part*.
- Previous recent laureates: Jon Fosse (2023), Annie Ernaux (2022), Abdulrazak Gurnah (2021).

⚠️ आयोग का TRAP / RATTA POINT

Rabindranath Tagore was the first Asian to win the Nobel Prize in Literature (1913); Han Kang is the first Asian WOMAN to win it!

Q. 220 Ancient Indian History | Harappan Seals and Pashupati Seal

Consider the following statements regarding the famous 'Pashupati Seal' discovered at Mohenjodaro:

1. It is a steatite seal depicting a three-faced male deity seated in a yogic posture (*Padmasana*) wearing a horned headdress.
2. Four wild animals surround the central deity: an Elephant and a Tiger on the right, and a Rhinoceros and a Buffalo on the left.
3. Two antelopes / deer are depicted beneath the stool/seat of the deity.

Which of the statements given above are correct?

- **A** 1 and 2 only
- **B** 2 and 3 only
- **C** 1 and 3 only
- **D** 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - Discovered in 1928–29 by Ernest Mackay at Mohenjodaro; identified by Sir John Marshall as *Proto-Shiva* (*Pashupati* - Lord of Beasts).
  - **Surrounding Animals (Total 5 species / 6 animals):**
  - Right side of deity: **Elephant** and **Tiger**.
  - Left side of deity: **Rhinoceros** and **Water Buffalo**.
  - Beneath the seat / at his feet: **Two Deer / Antelopes**.

⚠️ आयोग का TRAP / RATTA POINT

The **HORSE and COW are completely absent** from the Pashupati Seal (and Harappan seals in general)!

Q. 221 🌍 Geography of India 📌 Famous Waterfalls of Peninsular India

Match List-I (Waterfall) with List-II (River & State) and select the correct answer:

List-I (Waterfall)	List-II (River & State)
A Jog / Gersoppa Falls	1 Kaveri River; Karnataka
B Shivanasamudra Falls	2 Sharavathi River; Karnataka
C Dhuandhar Falls	3 Subarnarekha River; Jharkhand
D Hundru Falls	4 Narmada River; Madhya Pradesh

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Jog Falls (Gersoppa / Mahatma Gandhi Falls):** Plunge waterfall (253 m) on **Sharavathi River** in Shimoga, Karnataka; consists of four distinct cascades: Raja, Roarer, Rocket, and Rani.
- **Shivanasamudra Falls:** Segmented waterfall on **Kaveri River** in Chamarajanagar, Karnataka; site of India's first major hydroelectric power station (1902).
- **Dhuandhar Falls ("Smoky Falls"):** Located at Bhedaghat on **Narmada River** near Jabalpur, Madhya Pradesh, where the river plunges through marble rocks.
- **Hundru Falls:** Spectacular 98 m waterfall on **Subarnarekha River** near Ranchi, Jharkhand.

⚠️ आयोग का TRAP / RATTA POINT

Kunchikal Falls on Varahi River (Karnataka) is officially the highest tiered waterfall in India (~455 m)! Nohkalikai Falls (Meghalaya, 340 m) is the highest direct plunge waterfall.

Q. 222 🏛 Indian Polity 📌 74th Constitutional Amendment Act (Urban Local Bodies)

Consider the following statements regarding the 74th Constitutional Amendment Act, 1992:

1. It added Part IX-A to the Constitution titled "The Municipalities" and the Twelfth Schedule containing 18 functional items.
2. Under Article 243S, Ward Committees must be constituted in all municipalities having a population of three lakhs or more.
3. Under Article 243ZE, a Metropolitan Planning Committee must be constituted in every Metropolitan area to prepare a draft development plan.

Which of the statements given above are correct?

A 1 and 2 only

B 2 and 3 only

C 1 and 3 only

D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **74th CAA 1992 (effective 1 June 1993):** Constitutionalized Urban Local Bodies (Articles 243P to 243ZG); added **12th Schedule with 18 functional subjects** (urban planning, regulation of land use, roads, water supply, public health).
  - **Article 243S:** Mandatory constitution of **Wards Committees** in all municipalities with population $\ge 3,00,000$.
  - **Article 243ZD & 243ZE:** District Planning Committee (DPC) under 243ZD; Metropolitan Planning Committee (MPC) under 243ZE.

⚠️ आयोग का TRAP / RATTA POINT

11th Schedule (Panchayats) = **29 items**; 12th Schedule (Municipalities) = **18 items**!

Q. 223 🔬 General Science (Chemistry) 📌 Naturally Occurring Organic Acids

Match List-I (Natural Source) with List-II (Organic Acid Present) and select the correct answer:

List-I (Source)	List-II (Organic Acid)
A Ant sting & Nettle leaf sting	1 Acetic Acid (CH_3COOH)
B Vinegar	2 Formic Acid / Methanoic Acid ($HCOOH$)
C Tamarind & Unripe Grapes	3 Lactic Acid
D Sour Milk / Curd	4 Tartaric Acid

A A-2, B-1, C-4, D-3

B A-1, B-2, C-4, D-3

C A-2, B-4, C-1, D-3

D A-3, B-1, C-4, D-2

✓ Correct Answer :Option A

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- **Ant sting & Nettle:** Contains **Methanoic acid (Formic acid, $HCOOH$)**; causes burning pain, neutralized by applying a mild base like baking soda ($NaHCO_3$) or calamine lotion ($ZnCO_3$).
- **Vinegar:** 5–8% aqueous solution of **Acetic acid (CH_3COOH)**.
- **Tamarind & Grapes:** Rich in **Tartaric acid**.
- **Curd / Sour Milk:** Contains **Lactic acid** produced by *Lactobacillus* fermenting lactose.
- **Tomato & Spinach:** Contain **Oxalic acid** (excess intake can form calcium oxalate kidney stones).
- **Citrus fruits (lemon, orange):** Contain **Citric acid**.
- **Apple:** Contains **Malic acid**.

⚠️ आयोग का TRAP / RATTA POINT

Vitamin C is **Ascorbic Acid**, found abundantly in Amla (Indian gooseberry) and guava!

Q. 224 🎯 Uttar Pradesh Special 📌 History of Allahabad High Court

Consider the following statements regarding the High Court of Judicature at Allahabad:

1. It was originally established under the Indian High Courts Act, 1861, at Agra on June 11, 1866, as the High Court of Judicature for the North-Western Provinces.
2. The High Court was shifted from Agra to Allahabad in 1869.
3. Sir Walter Morgan was the first Chief Justice of the High Court.

Which of the statements given above are correct?

A 1 and 2 only

B 2 and 3 only

C 1 and 3 only

D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Foundation:** Established on June 11, 1866, at **Agra** (4th High Court in India after Calcutta, Bombay, Madras).
  - **Relocation:** Shifted to **Allahabad** in 1869; name officially changed to 'High Court of Judicature at Allahabad' in 1919.
  - **Chief Justices:** First Chief Justice was **Sir Walter Morgan** (1866–1871). First Indian Chief Justice was **Sir Shah Muhammad Sulaiman** (1932).
  - Under the UP High Court (Amalgamation) Order 1948, the Chief Court of Oudh was amalgamated into Allahabad High Court, retaining a permanent bench at **Lucknow**.

⚠️ आयोग का TRAP / RATTA POINT

Allahabad High Court has the LARGEST sanctioned judicial strength among all High Courts in India (160 judges)!

Q. 225 📜 Modern Indian History 📌 Subhas Chandra Bose and Azad Hind Fauj (INA)

Consider the following statements regarding Subhas Chandra Bose and the Indian National Army (INA):

1. Subhas Chandra Bose formed the 'Forward Bloc' within the Congress in May 1939 after resigning from the Congress presidency.
2. The Indian National Army (INA) was originally conceptualized and founded in Malaya by Captain Mohan Singh with Japanese Major Iwaichi Fujiwara.
3. Subhas Chandra Bose proclaimed the formation of the Provisional Government of Free India (*Arzi Hukumat-e-Azad Hind*) in Singapore on October 21, 1943.

Which of the statements given above are correct?

A 1 and 2 only	B 2 and 3 only
C 1 and 3 only	D 1, 2 and 3

✓ Correct Answer :Option D

📘 DETAILED SOLUTION & HIGH-YIELD ANALYSIS

- All three statements are correct:
  - **Forward Bloc:** Founded at Unnao/Makur (UP) in May 1939 after resigning following the Tripuri crisis with Gandhi and CWC.
  - **INA Genesis:** Conceived by **Captain Mohan Singh** in Malaya in 1941 from British-Indian prisoners of war; Rash Behari Bose organized the Indian Independence League in Tokyo and handed over supreme leadership to Subhas Bose in July 1943.
  - **Provisional Government:** Proclaimed on October 21, 1943, at Cathay Cinema in Singapore; recognized by 9 Axis powers. Japan handed over Andaman and Nicobar Islands to INA, renamed by Bose as **Shaheed Dweep** and **Swaraj Dweep**.

⚠️ आयोग का **TRAP / RATTA POINT**

Famous slogan "*Dilli Chalo*" and "*Tum Mujhe Khoon Do, Main Tumhe Azadi Doonga*" were given in Burma/Singapore! INA Red Fort trials (1945) defended by Bhulabhai Desai, Tej Bahadur Sapru, and Jawaharlal Nehru.